

000003



HYDROGEOLOGIC INVESTIGATION REPORT
FOR
FOUR STAR CORPORATION
CADILLAC, MICHIGAN

Prepared By:
GOSLING CZUBAK ASSOCIATES, P.C.
525 W. Fourteenth Street
Traverse City, Michigan 49684
616/946-9191

August 1983

EPA Region 5 Records Ctr.



234875

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SITE LOCATION AND DESCRIPTION

The site is located in the northerly portion of the City of Cadillac in an industrial park. Specifically, the site is located on Sixth Avenue between Seventh and Thirteenth Streets. The project location is shown in Figure 1. The former manufacturing plant is situated on approximately 11.5 acres and includes a 50,000 square foot building and 50,000 square feet of gravel parking lot. The site lies immediately west of the City of Cadillac's municipal well field.

Land use in the area is open space and industrial. The terrain is relatively flat. Ground cover generally consists of weeds and grasses.

A copy of an aerial photograph showing buildings in the vicinity and the location of the municipal well field is included in Appendix A. A site plan, including the location of soil borings and monitor wells, is included in Appendix B.

BACKGROUND INFORMATION

Four Star Corporation manufactured polished aluminum and stainless steel trim for the automobile industry at the Cadillac plant from 1965 until the plant was closed in 1979. The building has been vacant since that time. Four Star operated a vapor degreaser in its manufacturing process from 1978 to 1979; the degreaser utilized the solvent trichloroethylene (TCE). TCE fumes were vented outside the building in the location indicated on the site plan in Appendix B. It is thought that the venting of solvents and minor spills while handling solvent containers may have contaminated the soil and groundwater system beneath the site.

Four Star's Cadillac operations also used forming and rolling lubricants. Stained soil and gravel parking surfaces indicate that lubricants were disposed near the building or dripped off of scrap pieces stored in bins outside the building.

The building is served by municipal sewer and water service. The former plant manager is not aware of any tile field or septic system, and drilling activities did not encounter any tile field.

The industrial area in the vicinity of the former Four Star plant has several cases of documented chemical contamination of groundwater. At least two of these cases are directly attributable to industries other than Four Star. In addition to this study, other groundwater investigations are or have been conducted in the area.

The proximity of the Cadillac municipal well field and the documented presence of groundwater contamination is of major concern to City and State health officials. The Cadillac wells are 250' to 300' below the water table. Testing of the municipal water indicates that the wells are free of contamination and hydrogeologic investigations are being conducted by the City of Cadillac to determine the potential

GEOLOGIC SETTING

The study area is located in a glacial outwash plain between moraine systems to the north and south. The soil deposits are predominately clean sands with lenses of clay and silt. Surface water movement is to the east and southeast until it joins the Muskegon River in Clare County. Lakes Cadillac and Mitchell are located south of the project site and drain by surface and groundwater flow in a similar direction.

Gosling Czubak Associates



Engineers
Surveyors

Gosling Czubak Associates, P.C.
525 West Fourteenth
Traverse City, MI 49684
616 946-9191

PROJECT Four Star DATE STARTED 5/13/83
LOCATION Cadillac, Michigan DATE COMPLETED 5/13/83
CLIENT Ron Sondee DRILLER BRK HELPER BB & JCM

BORING NO. SB #7 Weather Clear - Breezy 70°F
Ground Surface Elev. 1294.5 Water Data 27.5'± below ground
Plugging Method Existing Material Job No. 83072.01

Depth	Sample	Sampling Method	PENETR.: Blow Count			LINER	Soil Classification
			1	2	3		
0.4							3"± ROAD GRAVEL
	SUR	SS	1	2	—		
5	SUR	SS	2	2	—		BROWN FINE 5
	SUR	SS	2	2	—		
	SUR	SS	2	3	—		TO MEDIUM 10
	SUR	SS	3	4	—		
15	SUR	SS	4	9	—		SAND 15
	SUR	SS	8	18	—		
20	SUR	SS	6	10	—		20
	SUR	SS	7	7	—		
25	SUR	SS	8	20	—		25
27.5	SUR	SS	10	14	—		
							END OF BORING 28.0 FT.
30							30
35							35

APPENDIX
LABRATORY ANYALYSIS RESULTS



5693 W. THIRD STREET
P.O. BOX 647
LUDINGTON, MICHIGAN 49431
PHONE 616-843-3372

JUN. 1 1983

Gosling Czubak Associates
525 West Fourteenth
Traverse City, MI. 49684
ATTN: Mr. Chuck Braumbaugh:

May 31, 1983

Dear Mr. Braumbaugh:

Please find below the results for the samples(soil) that were submitted to our laboratory for analysis on May 4, 1983.

ASI CODE: 83-060-4010 (EP Leachate)

P. LOT INFILTRATION
BASIN

PARAMETER	TRUCK WELL	PARKING LOT	POND	BACK DOOR	DEGREASER	BACK GROUND
<i>Selenium</i> Detection Limit = 0.005	ND	ND	ND	ND	0.01	ND
<i>Arsenic</i> Detection Limit = 0.02	ND	ND	ND	ND	ND	ND
<i>Silver</i> Detection Limit = 0.01	0.014	ND	ND	ND	ND	ND
<i>Chromium</i> Detection Limit = 0.01	0.05	0.05	0.03	0.03	0.05	0.01
<i>Lead</i> Detection Limit = 0.01	ND	0.03	0.13	ND	ND	ND
<i>Cadmium</i> Detection Limit = 0.01	0.079	ND	0.016	ND	ND	ND
<i>Mercury</i> Detection Limit = 0.001	0.002	ND	ND	ND	0.003	ND
<i>Barium</i> Detection Limit = 0.5	6.5	5.6	15.0	1.1	3.8	ND

ND = Non-detectable

Results and Detection limits are reported in mg/L

If you have any questions concerning this information, or if we may be of further assistance to you in the interest of Gosling Czubak Associates, please do not hesitate to contact our office.

Sincerely,
Aquatic Systems, Inc.

Bill dms
William L. Organ

ds

EPA #1310 EP LEACHATE



METHODS AND RESULTS OF CADILLAC FOUR STAR SOIL AND WATER ANALYSIS

Purpose: Soil and water samples from the Cadillac 4-Star site were submitted by Gosling Czubak Associates for a chlorinated solvent scan. Special attention was given to the detection and quantification of Trichloroethylene(TCE) and Perchloroethylene(PCE).

SAMPLE PREPARATION

Methods: For this project two types of samples were analyzed, soil and water. Soil samples were prepared using a modification of EPA method 5020. 20 grams of soil were sealed within a 40 ml glass serum vial which was then placed in a water bath at 90°C for one hour. One milliliter of the head space volume was then injected into the Gas Chromatograph. This method is generally referred to as Headspace Gas Analysis.

Water samples were prepared using EPA method 601. Five milliliters of water were injected into the purge chamber of a Tekmar LSC-1 where it was purged with nitrogen for eleven minutes. This method is generally referred to as Purge and Trap Analysis.

GAS CHROMATOGRAPH

Analysis: The GC was operated under the conditions suggested by EPA method 5030, with a Carbowack B/1% SP-1000 column and a carrier gas flow of nitrogen at 20 cc/minute. The column temperature was programmed to maintain a temperature of 50°C for three minutes, followed by an eight degree per minute increase in temperature until a temperature of 190°C was obtained. This 190°C temperature was maintained for four minutes. This method of detection was Flame Ionization (FIO).

Results: Most of the soils were found to have non-detectable concentrations of chlorinated solvents(e.g. MW #3 series). Some soil samples had detectable peaks and were designated as chlorinated solvents, in an estimated total amount (e.g. SB #1 at 12.5' chlorinated solvents >500 µg/L). A few of the soil samples displayed clearly identifiable peaks(e.g. SB #2 at 27.5', 650 µg/L TCE).



*NOTE: One sample from the second set of samples, SB #10 at 1.0', was heavily contaminated by high boiling organic compounds. The high temperature at which the peaks came off of the Gas Chromatograph and the fact that the contamination did not penetrate the soil, indicated that SB #10 may have been the site of an incidental grease or oil spill.



AQUATIC SYSTEMS, INC.
5693 West Third Street
PO Box 647
Ludington, MI. 49431
(616)843-3372

ASI CODE: 83-075-4015

MW#1
Dup.#1
MW#2
MW#3

ANALYSIS CONDUCTED FOR:

Gosling Czubak Associates/Cadillac Star

83-113-4015 MW#1, MW#4, MW#5 and MW#6

DEPTH	SAMPLE LOCATION	SAMPLE TYPE	ASI SAMPLE NUMBER	SAMPLE DATE	1,1-dichloro-ethane	1,1,1-trichloro-ethane	1,1,2,2-tetra-chloroethane	Perchloroethylene	Trichloroethylene	Chlorinated Solvents
FT				1983	µg/L	µg/L	µg/L	µg/L	µg/L	µg/L
40-43	MW#1	Water	0160	05-04	ND	620	ND	5000	21000	ND
--	Dup.1	Water	0257	05-04	ND	ND	ND	ND	ND	ND
40-43	MW#2	Water	0161	05-05	ND	ND	ND	ND	ND	ND
40-43	MW#3	Water	0162	05-05	ND	ND	ND	ND	ND	ND
--	MW#1	Water	0367	05-19	487	1158	ND	15342	59434	Present
--	MW#4	Water	0368	05-19	ND	307	ND	315	1151	ND
28-31	MW#5	Water	0369	05-13	110	334	ND	3934	22568	ND
28-31	MW#6	Water	0370	05-16	ND	ND	ND	20	41.5	ND

ND = Non-detectable

Detection Limit for Chlorinated Organics in the Water = 5 µg/L



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ANALYSIS CONDUCTED FOR:

Gosling Czubak Associates/Cadillac 4-Star

ASI CODE: 83-200-4015 MW#7, MW#8, MW#9, MW#10

DEPTH	SAMPLE LOCATION	SAMPLE TYPE	ASI SAMPLE NUMBER	SAMPLE DATE	1,1-dichloro-ethane	1,1,1-trichloro-ethane	1,1,2,2-tetra-chloroethane	Perchloroethylene	Trichloroethylene	Chlorinated Solvents
FT				1983	µg/L	µg/L	µg/L	µg/L	µg/L	µg/L
--	MW#7	Water	0601	07-05	ND	ND	ND	ND	11,344	ND
--	MW#8	Water	0600	07-05	ND	ND	ND	ND	3,047	ND
--	MW#9	Water	0599	07-05	ND	ND	ND	ND	69	ND
--	MW#10	Water	0602	07-05	ND	ND	ND	ND	ND	ND

ND = Non-detectable

Detection Limit for Chlorinated Organics in the water = 50 µg/L



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ASI CODE: 83-081 4015 MW#1
83-082-4015 MW#2

ANALYSIS CONDUCTED FOR:

Gosling Czubak Associates/Cadillac 4-Star

DEPTH	SAMPLE LOCATION	SAMPLE TYPE	ASI SAMPLE NUMBER	SAMPLE DATE	1,1-dichloro-ethane	1,1,1-trichloro-ethane	1,1,2,2-tetra-chloroethane	Perchloroethylene	Trichloroethylene	Chlorinated Solvents
FT				1983	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg
5	MW #1	Soil	0223	05-04	ND	ND	ND	ND	ND	ND
15			0225		ND	ND	ND	Trace	ND	ND
25			0227		ND	ND	ND	ND	ND	ND
35			0229		ND	ND	ND	ND	290	ND
45			0230		ND	ND	ND	ND	195	ND
60			0232		ND	ND	ND	ND	ND	ND
5	MW #2	Soil	0233	05-05	ND	ND	ND	ND	ND	ND
15			0235		ND	ND	ND	ND	ND	ND
40			0238		ND	ND	ND	ND	ND	ND

ND = Non-detectable

Detection Limit for Chlorinated Organics in the Soil = 30 µg/kg



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5693 West Third Street
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ASI CODE: 83-083-4015 MW#3
83-076-4015 SB#1

ANALYSTS CONDUCTED FOR:

Gosling Czubak Associates/Cadillac 4-Star

DEPTH	SAMPLE LOCATION	SAMPLE TYPE	ASI SAMPLE NUMBER	SAMPLE DATE	1,1-dichloro-ethane	1,1,1-trichloro-ethane	1,1,2,2-tetra-chloroethane	Perchloroethylene	Trichloroethylene	Chlorinated Solvents
FT				1983	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg
5	MW#3	Soil	0238	05-05	ND	ND	ND	ND	ND	ND
10			0239		ND	ND	ND	ND	ND	ND
20			0241		ND	ND	ND	ND	ND	ND
30			0243		ND	ND	ND	ND	ND	ND
40			0245		ND	ND	ND	ND	ND	ND
2.5	SB#1	Soil	0164	05-09	ND	ND	ND	ND	ND	ND
10			0166		ND	ND	ND	ND	ND	ND
12.5			0167		ND	ND	ND	ND	ND	>500
22.5			0171		ND	ND	ND	ND	ND	ND
30			0174		ND	ND	ND	ND	ND	>200

ND = Non-detectable

Detect Limit for Chlorinated Organics in the Soil = 30 µg/kg



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ASI CODE: 83-105-4 MW#4
83-106-4015 MW#5

ANALYSIS CONDUCTED FOR:

Gosling Czubak Associates/Cadillac 4-Star

DEPTH	SAMPLE LOCATION	SAMPLE TYPE	ASI SAMPLE NUMBER	SAMPLE DATE	1,1-dichloro-ethane	1,1,1-trichloro-ethane	1,1,2,2-tetra-chloroethane	Perchloroethylene	Trichloroethylene	Chlorinated Solvents
FT				1983	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg
2	MW#4	Soil	0288	05-12	ND	ND	ND	ND	ND	ND
12			0292		ND	ND	ND	ND	ND	ND
22			0296		ND	ND	ND	ND	ND	ND
25			0297		ND	ND	ND	ND	ND	ND
2	MW#5	Soil	0300	05-12	ND	ND	ND	ND	67	ND
12			0304		ND	ND	ND	ND	100	ND
22			0308		ND	ND	ND	245	381	Trace
25			0309		ND	ND	ND	Trace	315	ND

ND = Non-detectable

Detection Lim for Chlorinated Organics in the soil = 30 µg/kg



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(616)843-3372

ASI CODE: 83-07. 014 SB#2
83-078-4015 SB#3

ANALYSIS CONDUCTED FOR:

Gosling Czubak Associates/Cadillac 4-Star

DEPTH	SAMPLE LOCATION	SAMPLE TYPE	ASI SAMPLE NUMBER	SAMPLE DATE		1,1-dichloro-ethane		1,1,1-trichloro-ethane		1,1,2,2-tetra-chloroethane		Perchloroethylene		Trichloroethylene		Chlorinated Solvents
FT				1983		µg/kg		µg/kg		µg/kg		µg/kg		µg/kg		µg/kg
2.5	SB#2	Soil	0176	05-09		ND		ND		ND		ND		ND		ND
12.5			0180			ND		ND		ND		ND		ND		ND
20			0183			ND		ND		ND		ND		ND		ND
22.5			0184			ND		ND		ND		ND		ND		>200
27.5			0186			ND		ND		ND		ND		ND		ND
2.5	SB#3	Soil	0188	05-10		ND		ND		ND		ND		ND		ND
12.5			0192			ND		ND		ND		ND		ND		ND
22.5			0196			ND		ND		ND		ND		ND		ND

ND = Non-detectable

Detection Limit for Chlorinated Organics in the soil = 30 µg/kg



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(616)843-3372

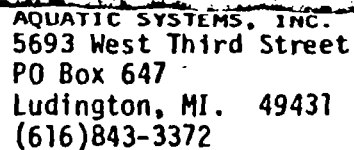
ASI CODE: 83-079 15 SB#4

Gosling Czubak Associates/Cadillac
4-Star

DEPTH	SAMPLE LOCATION	SAMPLE TYPE	ASI SAMPLE NUMBER	SAMPLE DATE	1,1-dichloro-ethane	1,1,1-trichloro-ethane	1,1,2,2-tetra-chloroethane	Perchloroethylene	Trichloroethylene	Chlorinated Solvents
FT				1983	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg
2.5	SB#4	Soil	0200	05-10	ND	720	ND	ND	ND	ND
5			0201		ND	500	ND	ND	ND	ND
7.5			0202		ND	ND	ND	ND	ND	ND
10			0203		ND	ND	ND	ND	ND	ND
12.5			0204		ND	ND	ND	ND	ND	ND
15			0205		ND	ND	ND	ND	ND	>500
17.5			0206		ND	ND	ND	ND	ND	ND
20			0207		ND	ND	ND	ND	ND	ND
22.5			0208		ND	ND	ND	ND	ND	ND
25			0209		ND	ND	ND	ND	ND	ND
27.5			0210		ND	ND	ND	ND	ND	>200

ND = Non-detectable

Detection Limit for Chlorinated Organics in the Soil = 30 µg/kg



ANALYSIS CONDUCTED FOR:

Gosling Czubak Associates/Cadillac 4-Star

[illegible]



AQUATIC SYSTEMS, INC.
5693 West Third Street
PO Box 647
Ludington, MI. 49431
(616) 843-3372

ASI CODE: 83-10. 4015 SB#6
83-108-4015 SB#7

ANALYSIS CONDUCTED FOR:

Gosling Czubak Associates/Cadillac 4-Star

DEPTH	SAMPLE LOCATION	SAMPLE TYPE	ASI SAMPLE NUMBER	SAMPLE DATE	1,1-dichloro-ethane	1,1,1-trichloro-ethane	1,1,2,2-tetra-chloroethane	Perchloroethylene	Trichloroethylene	Chlorinated Solvents
FT				1983	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg
1.5	SB#6	Soil	0310	05-13	ND	ND	ND	ND	ND	ND
2			0314		ND	ND	ND	895	ND	ND
2			0318		ND	ND	ND	ND	ND	ND
10			0321		ND	ND	ND	ND	78	ND
	SB#7	Soil	0322	05-13	ND	ND	ND	ND	ND	ND
2			0326		ND	ND	ND	ND	ND	ND
2			0330		ND	ND	ND	ND	ND	ND
7			0332		ND	ND	ND	ND	ND	ND

) = Non-detectable

Detection Limit for Chlorinated Organics in the Soil = 20 µg/kg



SYSTEMS, INC.
5093 West Third Street
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(616)843-3372

ASI CODE: 83-109-4015 SB#8
83-110-4015 SB#9

ANALYSIS CONDUCTED FOR:

Gosling Czubak Associates/Cadillac Star

DEPTH	SAMPLE LOCATION	SAMPLE TYPE	ASI SAMPLE NUMBER	SAMPLE DATE	1,1-dichloro-ethane	1,1,1-trichloro-ethane	1,1,2,2-tetra-chloroethane	Perchloroethylene	Trichloroethylene	Chlorinated Solvents
FT					µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg
2.5	SB#8	Soil	0333	05-13	ND	ND	ND	ND	ND	ND
12			0337		ND	ND	ND	ND	ND	ND
22			0341		ND	ND	ND	ND	ND	ND
27			0343		ND	ND	ND	ND	ND	ND
2	SB#9	Soil	0344	05-16	ND	ND	ND	460	ND	ND
15			0348		ND	ND	ND	550	ND	ND
25			0350		ND	ND	ND	420	ND	ND

ND = Non-detectable

Detection Limit for Chlorinated Organics in the soil = 30 µg/kg



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5690 West Third Street
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Ludington, MI. 49431
(616)843-3372

ASI CODE: 83-111-4015 SB#10
83-112-4015 SB#11

ANALYSIS CONDUCTED FOR:

Gosling Czubak Associates/Cadillac 4-Star

DEPTH	SAMPLE LOCATION	SAMPLE TYPE	ASI SAMPLE NUMBER	SAMPLE DATE	1,1-dichloro-ethane	1,1,1-trichloro-ethane	1,1,2,2-tetra-chloroethane	Perchloroethylene	Trichloroethylene	Chlorinated Solvents
FT				1983	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg	µg/kg
** 1	SB#10	Soil	0351	05-16	ND	ND	ND	ND	ND	ND
10			0355		ND	ND	ND	ND	ND	ND
20			0357		ND	ND	ND	ND	ND	ND
30			0359		ND	ND	ND	ND	ND	ND
2	SB#11	Soil	0360	05-16	ND	ND	ND	ND	ND	ND
15			0363		ND	ND	ND	ND	ND	ND
30			0366		ND	ND	ND	ND	ND	ND

ND= Non-detectable

Detection Limit for Chlorinated Organics in the soil

30 µg/kg ** see note enclosed

APPENDIX C
DNR CORRESPONDENCE



JAMES J. BLANCHARD, Governor

DEPARTMENT OF NATURAL RESOURCES

HOWARD A. TANNER, Director

CADILLAC DISTRICT HEADQUARTERS

RR #1, 8015 S. 131 Road

Cadillac, MI 49601

March 10, 1983

NATURAL RESOURCES COMMISSION

JOE A. HOEFER

LAITALA

MARY F. SNELL

PAUL H. WENDLER

HARRY H. WHITELEY

E.T. Peacy, V.P. Manufacturing
Four Star Corporation
1800 W. Maple
Troy, MI 48084

Re: Cadillac, MI Plant

Dear Mr. Peacy:

An inspection of the Cadillac plant and grounds was made with Pete Lukes of your Mesick plant on March 3, 1983.

A large area of stained soil was noted across the entire parking lot on the north side of the building. The stained area extended to a storm water seepage basin (trench) at the north edge of the parking lot. It was apparent that oily materials have been discharged or lost to the ground in that area.

A smaller area of stained soil and dead vegetation was noted outside a door on the west side of the plant.

A soil boring was made to 5½-6 feet with a hand auger in the seepage trench on the north side of the parking lot. Two soil samples were collected at approximately 5 ft and 5½ feet below the ground surface. The soil was sand which was clean in appearance but had a strong solvent odor. Photographs of the area on the north side of the plant were taken.

It is evident that operations at the Four Star Corp. plant in Cadillac have resulted in contamination of soil with oils and/or solvents. Such soil contamination may cause, or may already have caused, unlawful contamination of the groundwaters of the state. It is requested that Four Star Corporation immediately initiate corrective action as follows:

1. Define the extent of soil contamination at the site and submit a proposal for removal and proper disposal of contaminated soil as necessary.
2. Conduct a preliminary hydrogeologic study to determine if ground water has been contaminated as the result of the apparent soil contamination.

E. T. Peacy, V. P. Mfg.

-2-

March 10, 1983

Soil sampling and the hydrogeologic investigation should be done under the direction of a qualified groundwater geologist and in accordance with plans approved by this office.

Please advise this office of your intentions relative to this matter by no later than March 24, 1983. Contact this office if you have any questions.

Very truly yours,

GROUNDWATER QUALITY DIVISION

Daniel W. Darnell

Daniel W. Darnell
Water Quality Specialist

DWD:jb

cc: Gregg
Kincare
Weaver
Work

The static water levels were plotted and groundwater contours drawn to determine the direction of groundwater movement. This data has been placed on the map included in Appendix E. Examination of this map shows that groundwater enters the site flowing slightly east of due north and slowly turns to the north-northwest.

The cause of this shift in flow direction is not certain at this time. Several plausible explanations are under consideration. They include the following: (a) the relatively large area of roof and gravel parking is concentrating run-off infiltration in specific areas; (b) clay lenses near the surface are influencing percolation of rainfall; or (c) the Cadillac municipal well field is affecting groundwater patterns.

Throughout the project, there has been concern that the municipal well field is hydraulically connected to the upper aquifer. Hydraulic connection of these wells with the upper aquifer could influence the water level under the project site. Water levels in the monitor wells were accurately measured twice daily during the field investigation portion of this project to investigate this potential. The static water levels showed no variation between the morning and evening readings.

It can be seen from the map in Appendix E that the hydraulic gradient slope is approximately 0.17%. Based on estimates of soil permeability and porosity, the average groundwater flow velocity is about 50 to 75 feet/year.

SOIL AND GROUNDWATER CONTAMINATION:

Laboratory analysis of soil and groundwater samples were conducted. The purpose of the analysis was to:

- Determine the horizontal and vertical extents of any contamination
- Determine the composition of any contaminants
- Determine the concentration of any contaminants

TABLE 3

SOIL SAMPLE HYDROCARBON ANALYSIS RESULTS

ND = Nondetectable

All Results in ppm

Detection Limit = .030 ppm

Soil Boring	Depth (Ft.)	PARAMETER				
		1,1 Dichloro-ethane	1,1,1 Trichloro-ethane	1,1,2,2 Tetra-chloroethane	Perchloro-ethylene	Trichloro-ethylene
SB #2	2.5	ND	ND	ND	ND	ND
	12.5	ND	ND	ND	ND	ND
	20	ND	ND	ND	ND	ND
	22.5	ND	ND	ND	ND	.410
	27.5	ND	ND	ND	ND	.650
SB #3	2.5	ND	ND	ND	ND	ND
	12.5	ND	ND	ND	ND	ND
	22.5	ND	ND	ND	ND	ND
SB #4	2.5	ND	.720	ND	ND	ND
	5	ND	.500	ND	ND	ND
	7.5	ND	ND	ND	ND	ND
	10	ND	ND	ND	ND	ND
	12.5	ND	ND	ND	ND	ND
	15	ND	ND	ND	ND	ND
	17.5	ND	ND	ND	ND	ND
	20	ND	ND	ND	ND	ND
	22.5	ND	ND	ND	ND	ND
	25	ND	ND	ND	ND	ND
	27.5	ND	ND	ND	ND	ND

TABLE 3

SOIL SAMPLE HYDROCARBON ANALYSIS RESULTS

ND = Nondetectable

All Results in ppm

Detection Limit = .030 ppm

Soil Boring	Depth (Ft.)	PARAMETER				
		1,1 Dichloro-ethane	1,1,1 Trichloro-ethane	1,1,2,2 Tetra-chloroethane	Perchloro-ethylene	Trichloro-ethylene
SB #5	2.5	ND	ND	ND	ND	ND
	10	ND	ND	ND	ND	ND
	20	ND	ND	ND	ND	ND
	30	ND	ND	ND	Trace	Trace
MW #4	2	ND	ND	ND	ND	ND
	12	ND	ND	ND	ND	ND
	22	ND	ND	ND	ND	ND
	25	ND	ND	ND	ND	ND
MW #5	2	ND	ND	ND	ND	.067
	12	ND	ND	ND	ND	.100
	22	ND	ND	ND	.245	.381
	25	ND	ND	ND	Trace	.315
SB #6	2.5	ND	ND	ND	ND	ND
	12	ND	ND	ND	.895	ND
	22	ND	ND	ND	ND	ND
	30	ND	ND	ND	ND	.78
SB #7	2	ND	ND	ND	ND	ND
	12	ND	ND	ND	ND	ND
	22	ND	ND	ND	ND	ND
	27	ND	ND	ND	ND	ND

TABLE 3

SOIL SAMPLE HYDROCARBON ANALYSIS RESULTS

ND = Nondetectable

All Results in ppm

Detection Limit = .030 ppm

Soil Boring	Depth (Ft.)	PARAMETER				
		1,1 Dichloro-ethane	1,1,1 Trichloro-ethane	1,1,2,2 Tetra-chloroethane	Perchloro-ethylene	Trichloro-ethylene
SB #8	2.5	ND	ND	ND	ND	ND
	12	ND	ND	ND	ND	ND
	22	ND	ND	ND	ND	ND
	27	ND	ND	ND	ND	ND
SB #9	2	ND	ND	ND	.460	ND
	15	ND	ND	ND	.550	ND
	25	ND	ND	ND	.420	ND
SB #10	10	ND	ND	ND	ND	ND
	20	ND	ND	ND	ND	ND
	30	ND	ND	ND	ND	ND
SB #11	2	ND	ND	ND	ND	ND
	15	ND	ND	ND	ND	ND
	20	ND	ND	ND	ND	ND

TABLE 4

GROUNDWATER SAMPLE HYDROCARBON ANALYSIS RESULTS

Well No.	Sampling Date	PARAMETER				
		1,1 Dichloro- ethane	1,1,1 Trichloro- ethane	1,1,2,2 Tetra- chloroethane	Perchloro- ethylene	Trichloro- ethylene
1	5-4-83	ND	0.620	ND	5.000	21.000
1	5-19-83	0.487	1.158	ND	15.342	59.434
2	5-5-83	ND	ND	ND	ND	ND
3	5-5-83	ND	ND	ND	ND	ND
4	5-19-83	ND	0.307	ND	0.315	1.151
5	5-13-83	0.110	0.334	ND	3.934	22.568
6	5-16-83	ND	ND	ND	0.020	0.042
7	7-5-83	ND	ND	ND	ND	11.344
8	7-5-83	ND	ND	ND	ND	3.047
9	7-5-83	ND	ND	ND	ND	0.069
10	7-5-83	ND	ND	ND	ND	ND

ND = Nondetected

All Results in mg/l

Detection Limit = 0.005 mg/l

The data indicates that the groundwater has been contaminated with trichloroethylene (TCE) and perchloroethylene (PCE). It should be noted that the background quality of the groundwater (groundwater upstream of the project) from MW #6 contained some TCE and PCE, but in relatively minor levels when compared with other wells.

The contaminates are most concentrated in the truck loading area of the former plant (MW #5) and the parking lot drainage area (MW #1). The contaminate concentrations have been plotted on the map in Appendix E and an estimated contaminant plume configuration has been drawn. This map indicates that the plume is moving north off of the property of the former plant. The northern or leading edge of the contamination has not been located.

SUMMARY OF FINDINGS

The summary of findings from the hydrogeologic investigation, soil analysis and groundwater analysis is listed below:

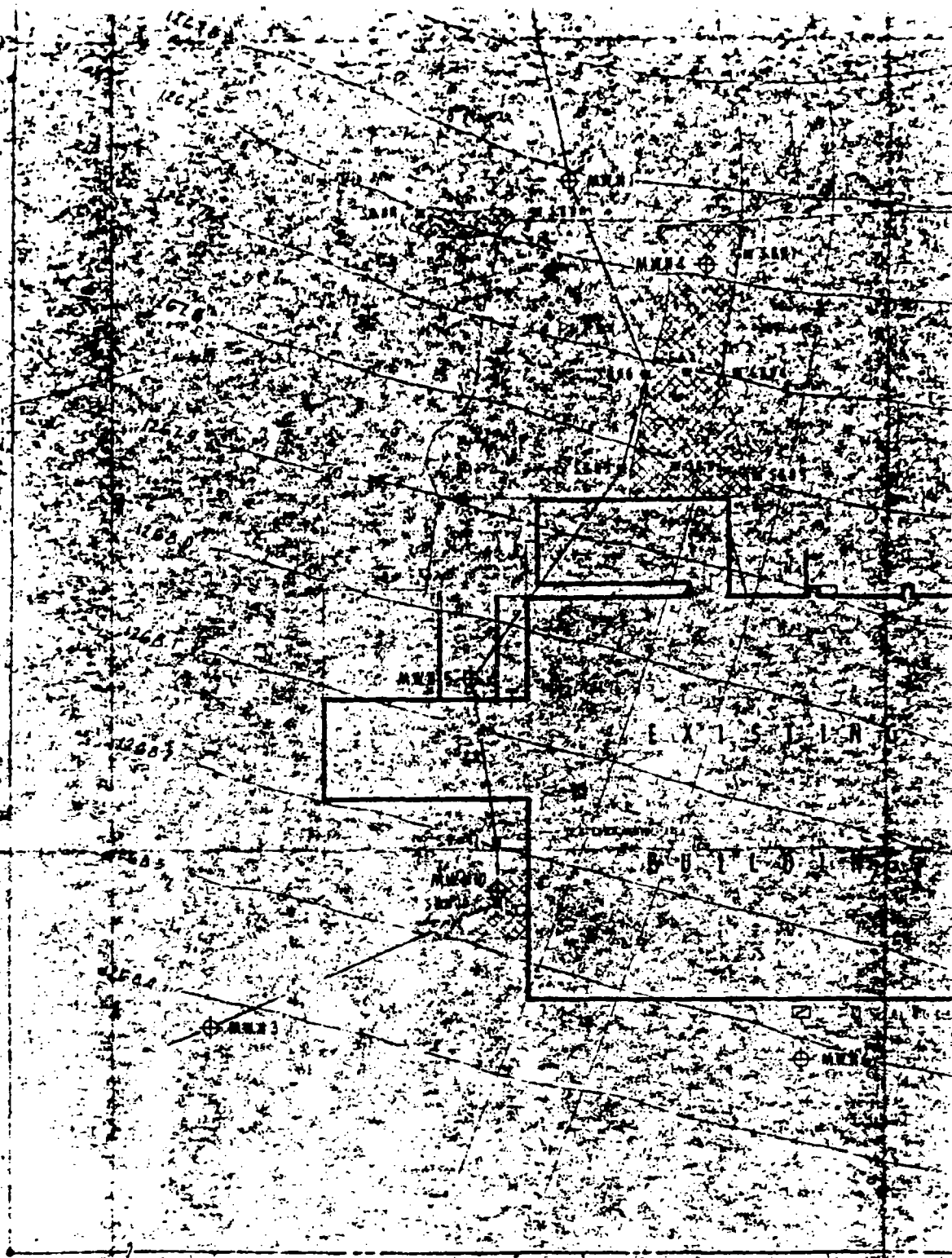
- The soils beneath the site are predominately sands;
- Groundwater occurs approximately 30 feet below land surface;
- Groundwater movement is generally north;
- Groundwater flow velocity is in the range of 50 to 75 ft./yr.;
- Some soils are contaminated with minor amounts of hydrocarbons;
- The stained soils do not contain significant concentrations of toxic metals;
- The groundwater is contaminated with chlorinated hydrocarbons, predominately TCE and PCE;
- Measurements of water levels in the monitor wells did not detect any influence of the upper aquifer by the Cadillac municipal wells;
- Some minor contamination is present in the background monitor well (MW #6);
- The leading edge of the contaminant plume has moved off of the property.

RECOMMENDATIONS

The following recommendations are offered:

- Establish a review meeting with the Department of Natural Resources to discuss the findings and future work;
- Receive permission to drill and install monitor wells from property owners north of the site. Identify the leading edge of the contaminant plume;
- Install several "cluster wells" to determine the vertical extent of the plume;
- Study should begin to determine remedial procedures.

APPENDIX A
AERIAL PHOTOGRAPH



Gosling Czubak Associates



Engineers
Surveyors

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Traverse City, MI 49684
616 946-9191

PROJECT Four Star
LOCATION Cadillac, Michigan
CLIENT Ron Sondee

DATE STARTED 5/13/83
DATE COMPLETED 5/13/83
DRILLER BRK HELPER BB & JCM

BORING NO. SB #6
Ground Surface Elev. 1295.5
Plugging Method Existing Material

Weather Clear - Breezy 60°F
Water Data 29½ ft± below ground
Job No. 83072.01

Depth	Sample	Sampling Method	PENETR.: Blow Count			LINER	Soil Classification
			1	2	3		
0.4							<u>3"± ROAD GRAVEL</u>
	<u>SNR</u>	<u>SS</u>	<u>4</u>	<u>4</u>	<u>—</u>		
5	<u>SNR</u>	<u>SS</u>	<u>3</u>	<u>3</u>	<u>—</u>		<u>BROWN FINE</u>
	<u>SNR</u>	<u>SS</u>	<u>2</u>	<u>2</u>	<u>—</u>		<u>TO MEDIUM</u>
10	<u>SNR</u>	<u>SS</u>	<u>2</u>	<u>3</u>	<u>—</u>		<u>SAND</u>
	<u>SNR</u>	<u>SS</u>	<u>2</u>	<u>5</u>	<u>—</u>		
15	<u>SNR</u>	<u>SS</u>	<u>4</u>	<u>5</u>	<u>—</u>		
	<u>SNR</u>	<u>SS</u>	<u>3</u>	<u>8</u>	<u>—</u>		
20	<u>SNR</u>	<u>SS</u>	<u>0</u>	<u>18</u>	<u>—</u>		<u>BROWN FINE TO MEDIUM WITH</u>
21.0	<u>SNR</u>	<u>SS</u>	<u>9</u>	<u>23</u>	<u>—</u>		<u>ALTERNATING GRAVEL LAYERS</u>
25	<u>SNR</u>	<u>SS</u>	<u>5</u>	<u>14</u>	<u>—</u>		<u>BROWN FINE</u>
	<u>SNR</u>	<u>SS</u>	<u>8</u>	<u>9</u>	<u>—</u>		<u>TO MEDIUM</u>
29.5							<u>SAND</u>
30	<u>SNR</u>	<u>SS</u>	<u>7</u>	<u>8</u>	<u>—</u>		<u>RED CLAY</u>
							<u>END OF BORING 30.5 FT.</u>
35							

HYDROGEOLOGIC INVESTIGATION REPORT

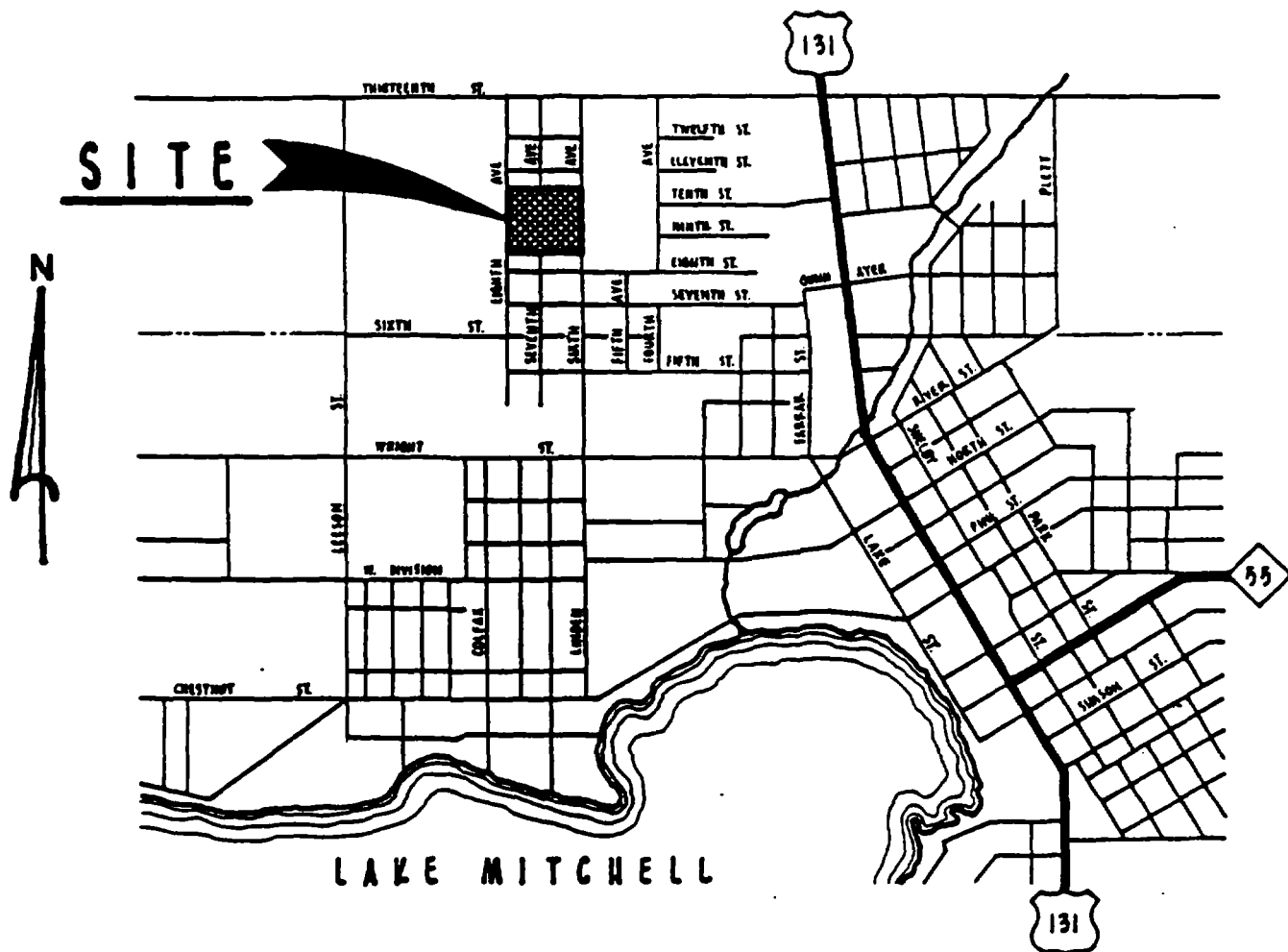
FOR

FOUR STAR CORPORATION

CADILLAC, MICHIGAN

INTRODUCTION AND PURPOSE

This report presents the findings of a hydrogeologic investigation performed at the Four Star Corporation's former manufacturing facility in the City of Cadillac, Michigan. The purpose of the investigation was to determine if soils and groundwater have been chemically contaminated and to define the extent of any contamination. The work has been performed at the request of the Michigan Department of Natural Resources.



FOUR STAR CORPORATION FORMER CADILLAC PLANT

LOCATION MAP NO SCALE

FIGURE 1

for future contamination of these wells.

On March 3, 1983, a Michigan Department of Natural Resources official inspected the site and made several shallow hand auger borings. Soil samples from the soil boring in the seepage area for parking lot runoff had a solvent odor. Subsequently, the Department of Natural Resources (DNR) requested that the soil contamination be defined and properly disposed and that a hydrogeologic study be conducted to determine if any groundwater contamination had occurred.

A copy of correspondence from the DNR to Four Star is included in Appendix C.

METHOD OF ANALYSIS

SOILS INVESTIGATION:

A soils investigation was conducted to define the horizontal and vertical extent of the contaminated soils. A series of eleven (11) soil borings were made in the locations shown on the site plan (SB #1 to SB #11). The soil borings averaged 28 to 31 feet in depth. The soil boring logs are included in Appendix D.

Borings were made using hollow stem augers. Soil samples were obtained at 2½ or 5 foot intervals by driving a split barrel sampler into the soil below the bottom of the augers. Augers, drilling tools and sampling equipment were steam cleaned at the completion of each hole. Sampling equipment was rinsed with technical grade methanol between samples.

Soil samples removed from the split barrel sampler were classified and logged and representative portions preserved in glass jars. The soil boring logs also noted soil types, strata changes and ground-water data.

GROUNDWATER INVESTIGATION:

A groundwater investigation was made to determine the occurrence of groundwater, depth to groundwater, direction of groundwater movement and water quality. A total of ten (10) monitor wells (MW #1 to MW #10) was installed in the locations shown on the plans. Monitor well logs are included in Appendix D.

The monitor wells have two-inch diameter casings and stainless steel screens. The screen length is 24" to 30", excepting MW #4 which has a ten foot long screen bisecting the water table. The screen depth

below the water table varies between 5 to 10 feet. All well casing materials, screens and fittings were steam cleaned prior to installation.

Soil samples were collected at 2½ to 5 foot intervals during drilling for the monitor wells. The soil samples were classified and stored as previously discussed. The drilling for some of the monitor wells was advanced an average of 20 to 30 feet below the static water level to log soil strata beneath the well screens. Wells were pumped and water samples were collected for analysis.

LABORATORY ANALYSIS:

Laboratory analysis was performed on soil and groundwater samples. The purpose of the analysis was to determine the existence and nature of any contamination. Groundwater samples were analyzed for chlorinated solvents. Soil samples were analyzed for both chlorinated solvents and other toxic materials that may have been contained in cutting oils.

Soil samples were prepared for chlorinated solvent analysis using a modification of EPA method 5020. Twenty (20) grams of soil were sealed within a 40 ml glass serum vial which was then placed in a water bath at 90°C for one hour. One milliliter of the head space volume was then injected into the Gas Chromatograph. This method is generally referred to as Headspace Gas Analysis.

Water samples were prepared for solvent analysis using EPA method 601. Five milliliters of water were injected into the purge chamber of a Tekmar LSC-1 where it was purged with nitrogen for eleven minutes. This method is generally referred to as Purge and Trap Analysis.

The sample analysis was performed in a Gas Chromatograph (GC). The GC performed a chlorinated solvent scan with special attention given to the detection and quantification of trichloroethylene (TCE) and

perchloroethylene (PCE).

The GC was operated under the conditions suggested by EPA method 5030 with a Carbopack B/1% SP-1000 column and carrier gas flow of nitrogen at 20 cc/minute. The column temperature was programmed to maintain a temperature of 50°C for three minutes, followed by an eight degree per minute increase in temperature until a temperature of 190°C was obtained. This 190°C temperature was maintained for four minutes. This method of detection is referred to as Flame Ionization (FIO).

Soil samples taken below the areas of stained soil were analyzed for toxic material content using the EPA method 1310 EP Leachate Procedure.

FINDINGS

SOIL CONDITIONS:

The soil borings revealed that the subsurface soils are fine to medium sands with some coarser material in variable amounts. Thin lenses of grey sandy clay and brown sandy clay were found throughout the site. The clay lenses do not appear to be continuous. A generalized north-south soil profile is shown on the site plan in Appendix B. Logs of the soil borings are included in Appendix D.

GROUNDWATER CONDITIONS:

The monitor wells show that groundwater exists at an average depth of 30 feet below land surface. The aquifer is contained in a sandy formation that is easily pumped.

Elevations were run on the top of each monitor well and depth to static water level was measured. A summary of this data is shown in Table 1. The top of casing and water level is referenced to USGS Datum.

TABLE I
GROUNDWATER DATA SUMMARY

<u>Monitor Well No.</u>	<u>Top Casing Elevation</u>	<u>Casing Length (ft.)</u>	<u>Screen Elevation</u>		<u>Static Water Level (7-15-83)</u>
			<u>Top</u>	<u>Bottom</u>	
1	1300.69	42	1256.2	1258.7	- *
2	1300.11	42	1255.6	1258.1	1267.45
3	1297.35	37.25	1260.1	1257.6	1268.44
4	1296.86	24.5	1262.4	1272.4	1267.50
5	1294.89	31.5	1260.9	1263.4	1267.65
6	1297.78	34.5	1260.8	1263.3	1268.32
7	1301.39	42	1257.4	1259.4	1267.24
8	1301.45	42	1257.5	1259.5	1267.25
9	1301.20	42	1257.2	1259.2	1267.28
10	1299.62	35.5	1262.1	1264.1	1268.08

*Static water level data is not available from MW #1, because vandals have rendered the well unusable.

- Determine if any soil contamination could leach groundwater in the future

The test methods for the analysis have been previous

The results of the laboratory analysis are included in .

Soil

Analysis was made for two general types of soil contamination. Samples from five (5) areas having stained soils were collected at the surface; an EP Leachate test was performed to characterize the waste from these samples. The test results show that no significant amount of metals exists and therefore poses no threat to groundwater. The results of the EP Leachate tests are summarized in Table 2.

TABLE 2

SOIL SAMPLE EP LEACHATE TEST RESULTS

<u>Parameter</u>	<u>Detection Limit</u>	<u>Truck Well</u>	<u>SAMPLING POINT</u>			<u>Degreaser Vent</u>	<u>Back- Ground</u>
			<u>Parking Lot</u>	<u>Leaching Basin</u>	<u>Back Door</u>		
Selenium	0.005	ND	ND	ND	ND	0.01	ND
Arsenic	0.02	ND	ND	ND	ND	ND	ND
Silver	0.01	0.014	ND	ND	ND	ND	ND
Chromium	0.01	0.05	0.05	0.03	0.03	0.05	0.01
Lead	0.01	ND	0.03	0.13	ND	ND	ND
Cadmium	0.01	0.079	ND	0.016	ND	ND	ND
Mercury	0.001	0.002	ND	ND	ND	0.003	ND
Barium	0.5	6.5	5.6	15.0	1.1	3.8	ND

ND = Nondetectable

All results in mg/l

The second portion of analysis was made on the soil samples recovered from soil borings made in the truck well, parking lot area and near the degreaser venting area. These samples were specifically analyzed for chlorinated hydrocarbon content.

The results of the lab analysis show some relatively low levels of contamination in the soil. The compounds that were discovered varied in concentration from one location to another. No pattern of soil contamination is evident. Furthermore, there is no indication that soil conditions have stratified contaminate concentrations. The results of this analysis are summarized in Table 3.

Groundwater

The groundwater quality beneath the project site was determined by analysis of water samples taken from the monitor wells. Samples were obtained by pumping the well for a sufficient time to insure that the sample was representative of existing conditions. Water samples were collected in 40 ml glass vials and capped with teflon closures; they were then refrigerated until analyzed. Analysis for chlorinated hydrocarbons was conducted according to the earlier described procedures.

The results of the groundwater analysis are summarized in Table 4.

TABLE 3

SOIL SAMPLE HYDROCARBON ANALYSIS RESULTS

ND = Nondetectable

All Results in ppm

Detection Limit = .030 ppm

Soil Boring	Depth (Ft.)	PARAMETER				
		1,1 Dichloro-ethane	1,1,1 Trichloro-ethane	1,1,2,2 Tetra-chloroethane	Perchloro-ethylene	Trichloro-ethylene
MW #1	5	ND	ND	ND	ND	ND
	15	ND	ND	ND	Trace	ND
	25	ND	ND	ND	ND	ND
	35	ND	ND	ND	ND	ND
	45	ND	ND	ND	ND	ND
	60	ND	ND	ND	ND	ND
MW #2	5	ND	ND	ND	ND	ND
	15	ND	ND	ND	ND	ND
	40	ND	ND	ND	ND	ND
MW #3	-	-	-	-	-	-
	25	ND	ND	ND	ND	ND
	27.5	ND	ND	ND	ND	ND

APPENDIX D
SOIL BORING LOGS
AND
MONITOR WELL LOGS

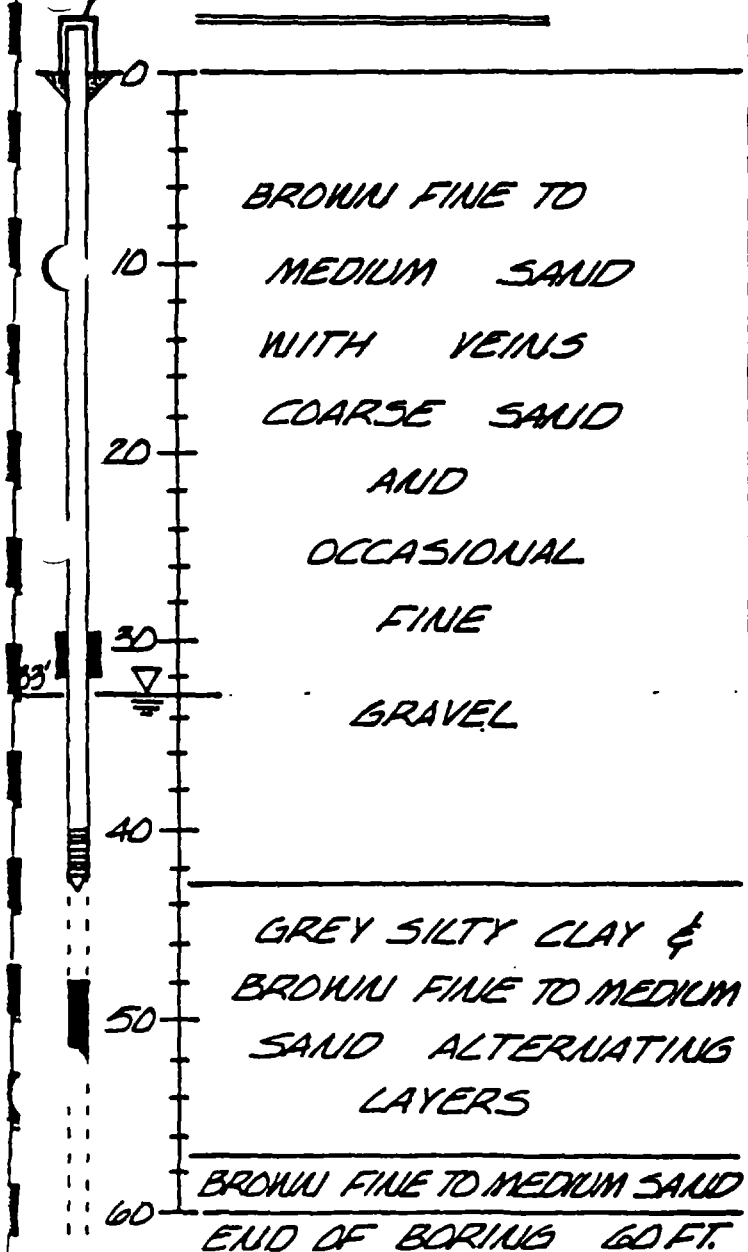


MONITOR WELL LOG

Project FOUR STAR
Location CADILLAC, MICHIGAN
Client RON SONDEE
Well Designation M.W. #1
Date Installed MAY 4, 1983

Development Method PUMPING
Ground Elevation 1298.4
Top of Casing Elev. 1300.69
Driller BRK Helper JCM, CDB
Job Number 83072.01

CASING PROTECTION SOIL FORMATION DATA



CASING DATA

Type 2" GALVANIZED
Length 42 FT.

SCREEN DATA

Type STAINLESS STEEL GAUZE
Size 2" DIA., 60 GAUGE
Length 30"

SAMPLING RECORD

Date MAY 4, 1983
Comments DUMPED W/ JET

■ BENTONITE SEAL
■ CONCRETE

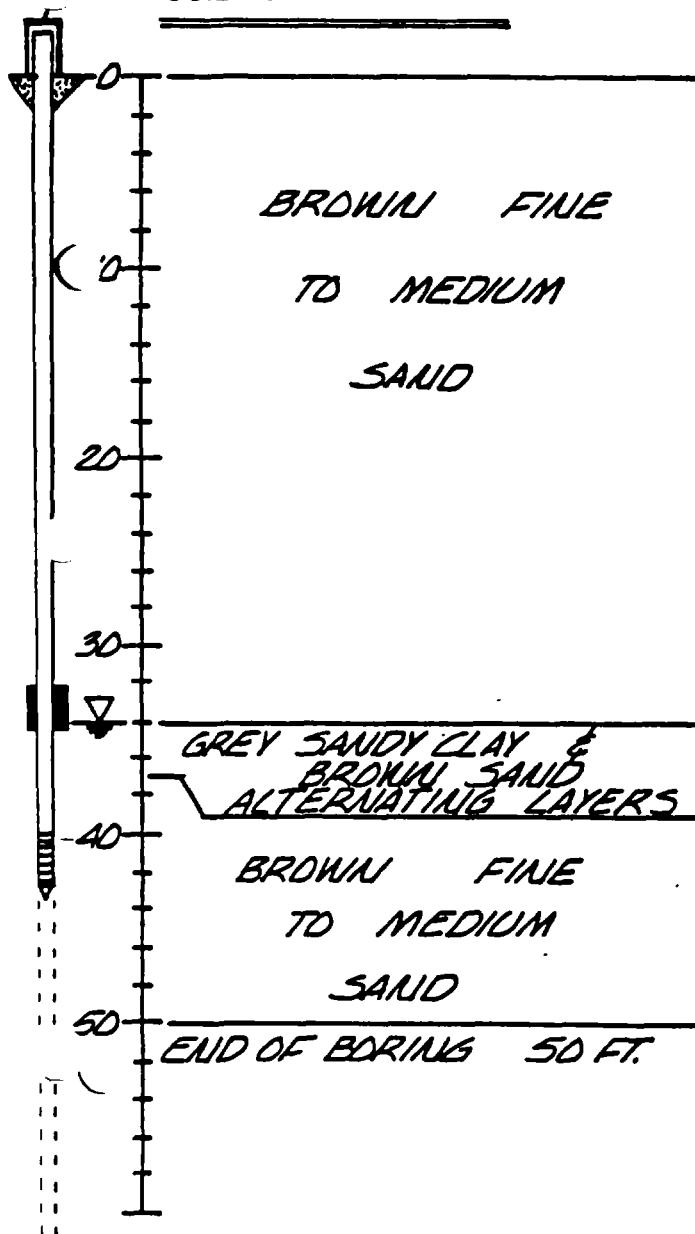


MONITOR WELL LOG

Project FOUR STAR
Location CADILLAC MICHIGAN
Client RON SONDEE
Well Designation M.W. #2
Date Installed MAY 5, 1983

Development Method PUMPING
Ground Elevation 1298.2
Top of Casing Elev. 1300.11
Driller BRK Helper JCM, CDB
Job Number 83072.01

CASING PROTECTION SOIL FORMATION DATA



CASING DATA

Type 2" GALVANIZED STEEL
Length 42 FT.

SCREEN DATA

Type STAINLESS STEEL GAUZE
Size 2" DIA., 60 GAUGE
Length 30"

SAMPLING RECORD

Date MAY 5, 1983
Comments PUMPED & SAMPLED
AT 2:45 P.M., TOOK DUPLICATE
SAMPLE #1 AS LAB. CHECK.

■ BENTONITE SEAL
▨ CONCRETE

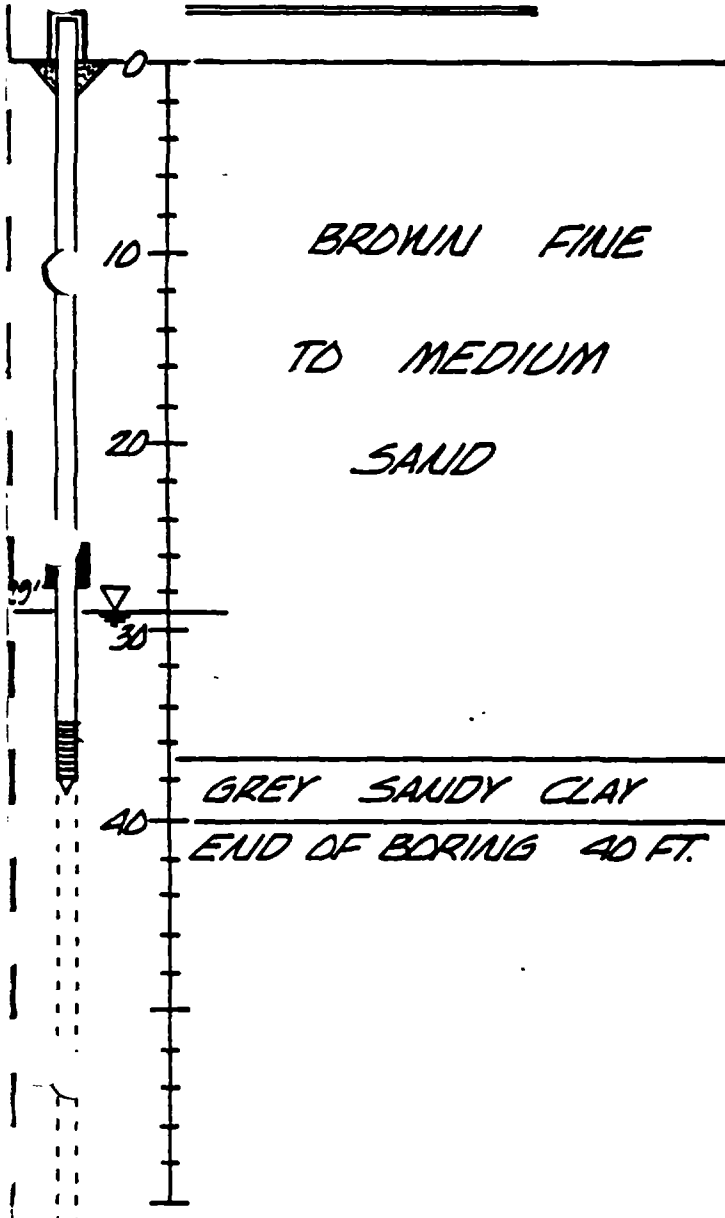


MONITOR WELL LOG

Project FOUR STAR
Location CADILLAC, MICHIGAN
Client RON SAUDEE
Well Designation M.W. #3
Date Installed MAY 5, 1983

Development Method PUMPING
Ground Elevation 1295.2
Top of Casing Elev. 1297.35
Driller BRK Helper JCM, CDB
Job Number 83072.01

CASING PROTECTION SOIL FORMATION DATA



CASING DATA

Type 2" DIA. GALVANIZED STEEL
Length 37.3 FT.

SCREEN DATA

Type STAINLESS STEEL GAUZE
Size 2" DIA., 60 GAUZE
Length 30"

SAMPLING RECORD

Date MAY 6, 1983
Comments WELL WAS DEVELOPED
AND SAMPLED WITH A JET
PUMP.

■ BENTONITE SEAL
■ CONCRETE

Gosling Czubak Associates



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616 946-9191

PROJECT Four Star
LOCATION Cadillac, Michigan
CLIENT Ron Sondee

DATE STARTED 5/9/83
DATE COMPLETED 5/9/83
DRILLER BRK HELPER CDB

BORING NO. S.B. #2
Ground Surface Elev. 1295.6
Plugging Method Exc. Soil

Weather Clear & Warm
Water Data 28'±
Job No. 83072.01

Depth	Sample	Sampling Method	PENETR.: Blow Count			LINER	Soil Classification
			1	2	3		
0.5							ROAD GRAVEL (BLACK TO 0.3')
	1	SS	2	4	—		LIGHT BROWN
5	2	SS	3	4	—		5 FINE TO MEDIUM
	3	SS	3	4	—		
10	4	SS	2	3	—		10 SAND
	5	SS	1	3	—		WITH A TRACE
15	6	SS	2	4	—		15 MEDIUM TO COARSE
	7	SS	2	4	—		
20	8	SS	3	5	—		20 SAND AND
	9	SS	4	6	—		FINE
25	10	SS	3	4	4		25 GRAVEL
28.0	11	SS	4	7	8		
30							END OF BORING 28.5 FT.
35							35

Gosling Czubak Associates



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Surveyors

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PROJECT Four Star DATE STARTED 5/9/83
LOCATION Cadillac DATE COMPLETED 5/9/83
CLIENT Ron Sondee DRILLER CDB HELPER BRK

BORING NO. S.B. #1 Weather Clear & Sunny
Ground Surface Elev. 1296.9 Water Data 29.0'±
Plugging Method Exc. Soil Job No. 83072.01

Depth	Sample	Sampling Method	PENETR.: Blow Count			LINER	Soil Classification
			1	2	3		
1.0							<i>BLACK SAND & GRAVEL</i>
	1	SS	6	8	—		<i>LIGHT BROWN FINE TO MEDIUM SAND</i>
5	2	SS	2	3	—		<i>BROWN SANDY CLAY</i>
6.0							
	3	SS	3	5	—		
10	4	SS	3	5	—		<i>LIGHT BROWN</i>
	5	SS	3	4	—		<i>FINE TO MEDIUM</i>
15	6	SS	4	7	—		<i>SAND</i>
	7	SS	2	3	—		<i>WITH LENSES</i>
20	8	SS	4	8	—		<i>OF</i>
	9	SS	10	10	—		<i>MEDIUM TO COARSE</i>
25	10	SS	5	12	—		<i>SAND AND</i>
	11	SS	8	14	—		
29.0							
30	12	SS	11	23	—		<i>FINE GRAVEL</i>
							<i>END OF BORING 30.5 FT.</i>
35							

Gosling Czubak Associates



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PROJECT Four Star
LOCATION Cadillac, Michigan
CLIENT Ron Sondee

DATE STARTED 5/10/83
DATE COMPLETED 5/10/83
DRILLER BRK HELPER JCM/CDB

BORING NO. SB #4
Ground Surface Elev. 1295.5
Plugging Method Existing Material

Weather Clear & Calm 55°F
Water Data 28'± below ground
Job No. 83072.01

Depth	Sample	Sampling Method	PENETR.: Blow Count			LINER	Soil Classification
			1	2	3		
0.4							3" ROAD GRAVEL
5	1	SS	5	5	—		BROWN
	2	SS	2	4	—		
	3	SS	2	4	—		
10	4	SS	2	4	—		FINE TO
	5	SS	3	5	—		
15	6	SS	2	5	—		MEDIUM
	7	SS	7	9	—		
20	8	SS	3	5	—		SAND
	9	SS	5	7	—		
25	10	SS	7	13	—		
28.0	11	SS	13	18	24		END OF BORING 28.0 FT.
30							
35							

Gosling Czubak Associates



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PROJECT Four Star
LOCATION Cadillac, Michigan
CLIENT Ron Sondee

DATE STARTED 5/10/83
DATE COMPLETED 5/10/83
DRILLER BRK HELPER JCM

BORING NO. SB #3
Ground Surface Elev. 1296.7
Plugging Method Existing Material

Weather Clear - Calm 45°F
Water Data 29'± below ground
Job No. 83072.01

Depth	Sample	Sampling Method	PENETR.: Blow Count			LINER	Soil Classification
			1	2	3		
0.25							ROAD GRAVEL
5	1	SS	3	4	—		BROWN
	2	SS	3	5	—		
	3	SS	3	5	—		
10	4	SS	2	3	—		FINE
	5	SS	3	4	—		
15	6	SS	3	5	—		TO
	7	SS	3	4	—		
20	8	SS	7	9	—		MEDIUM
	9	SS	11	17	—		
25	10	SS	13	22	—		SAND
	11	SS	14	25	—		
29.0	12	SS	14	22	—		
30							END OF BORING 30.5 FT.
35							

Gosling Czubak Associates



Engineers
Surveyors

Gosling Czubak Associates, P.C.
525 West Fourteenth
Traverse City, MI 49684
616 946-9191

PROJECT Four Star
LOCATION Cadillac, Michigan
CLIENT Ron Sondee

DATE STARTED 5/10/83
DATE COMPLETED 5/10/83
DRILLER BRK HELPER JCM/CDB

BORING NO. SB #5
Ground Surface Elev. 1296.9
Plugging Method Existing Material

Weather Clear & Calm 65°F
Water Data 29'± below ground
Job No. 83072.01

Depth	Sample	Sampling Method	PENETR.: Blow Count			LINER	Soil Classification
			1	2	3		
0.4							3" ROAD GRAVEL
1	1	SS	3	4	—		
5	2	SS	2	2	—		BROWN
	3	SS	3	6	—		
10	4	SS	2	4	—		FINE TO
	5	SS	3	5	—		
15	6	SS	3	5	—		MEDIUM
	7	SS	3	5	—		
20	8	SS	4	6	—		SAND
	9	SS	2	4	—		
25	10	SS	6	10	—		
	11	SS	8	8	—		
29.0							
30	12	SS	3	5	7		
							END OF BORING 31.0 FT.
35							

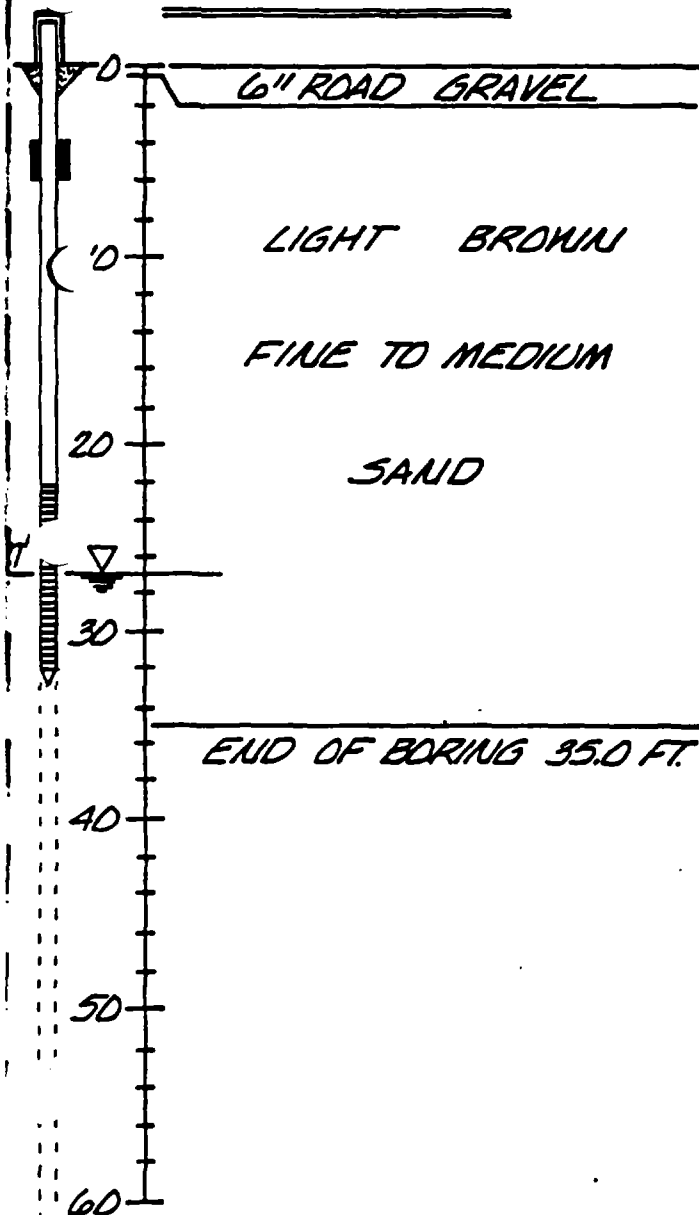


MONITOR WELL LOG

Project FOUR STAR
Location CADILLAC, MICHIGAN
Client RON SONDEE
Well Designation MW#4
Date Installed MAY 12, 1983

Development Method BAILING
Ground Elevation 1294.6
Top of Casing Elev. 1296.86
Driller BRK Helper BJB
Job Number 83073.01

CASING PROTECTION SOIL FORMATION DATA



CASING DATA

Type 2\" GALVANIZED
Length 24.5 FT.

SCREEN DATA

Type STAINLESS STEEL MESH
Size #60 GAUGE, 2\" DIA.
Length 120\" (10.0 FT.)

SAMPLING RECORD

Date _____
Comments BAILED BY HAND

 BENTONITE SEAL
 CONCRETE

Gosling Czubak

Associates



Engineers
Surveyors

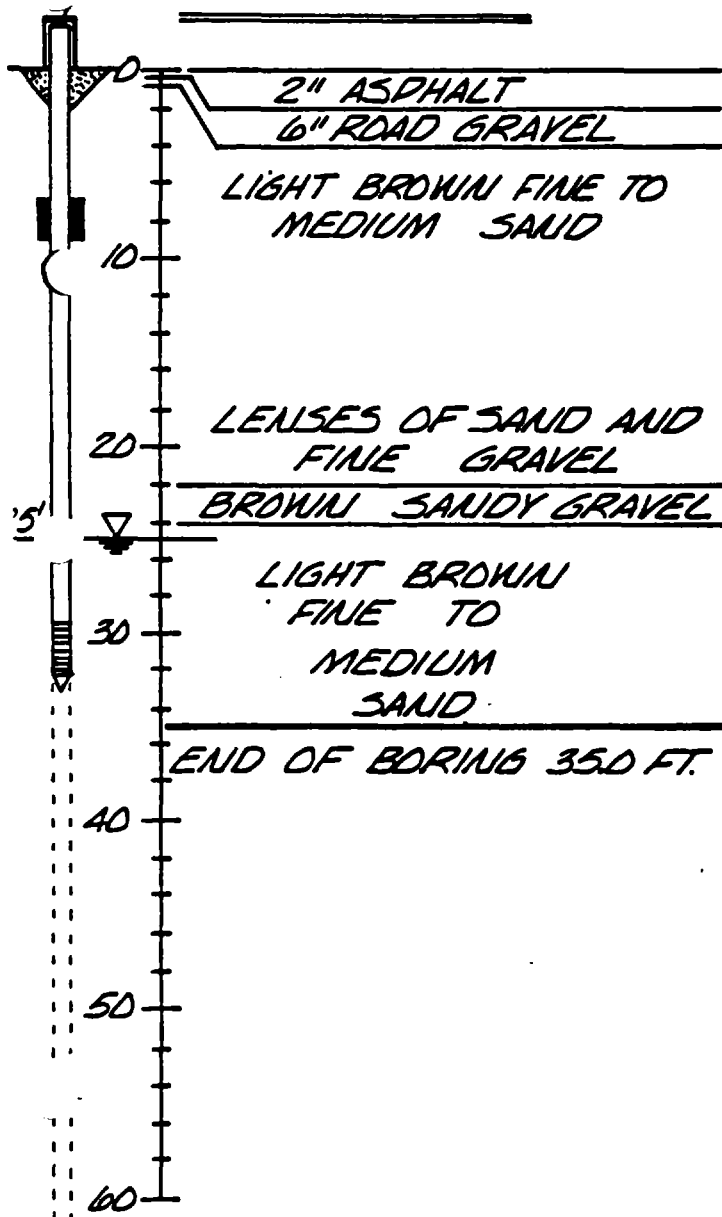
Gosling Czubak Associates, P.C.
525 West Fourteenth
Traverse City, MI 49684
616 946-9191

MONITOR WELL LOG

Project FOUR STAR
Location CADILLAC, MICHIGAN
Client RON SONDEE
Well Designation M.W.#5
Date Installed MAY 12, 1983

Development Method PUMPING
Ground Elevation 1293.3
Top of Casing Elev. 1294.89
Driller BRK Helper RB/CDB
Job Number 83072.01

CASING PROTECTION SOIL FORMATION DATA



CASING DATA

Type 2\" GALVANIZED
Length 31.5 FT.

SCREEN DATA

Type STAINLESS STEEL GAUZE
Size 2\" DIA., 60 GAUGE
Length 30\"

SAMPLING RECORD

Date MAY 12, 1983
Comments PUMPED W/ JET

 BENTONITE SEAL
 CONCRETE

Gosling Czubak Associates, P.C.
525 West Fourteenth
Traverse City, MI 49684
616 946-9191

DATE STARTED 5/13/83
DATE COMPLETED 5/13/83
DRILLER BRK HELPER BB & JCM

Weather Clear & Breezy 75°F
Water Data 27'± below ground
Job No. 83072.01

Depth	Sample	Sampling Method	PENETR.: Blow Count			LINER	Soil Classification
			1	2	3		
	SNR	SS	2	2	—		
5	SNR	SS	2	2	—		BROWN FINE
	SNR	SS	2	3	—		
10	SNR	SS	4	9	—		TO MEDIUM
	SNR	SS	5	6	—		
15	SNR	SS	6	8	—		SAND
	SNR	SS	8	20	—		
20	SNR	SS	9	9	—		
	SNR	SS	6	8	—		
25	SNR	SS	10	20	—		
27'	SNR	SS	8	11	—		
							END OF BORING 28.0 FT.
30							
35							

Gosling Czubak Associates



Engineers
Surveyors

Gosling Czubak Associates, P.C.
525 West Fourteenth
Traverse City, MI 49684
616 946-9191

PROJECT Four Star
LOCATION Cadillac, Michigan
CLIENT Ron Sondee

DATE STARTED 5/16/83
DATE COMPLETED 5/16/83
DRILLER BRK HELPER BB & JCM

BORING NO. SB #9
Ground Surface Elev. 1292.7
Plugging Method Existing Material

Weather Cloudy & Calm 40°F
Water Data 25'± below ground
Job No. 83072.01

Depth	Sample	Sampling Method	PENETR.: Blow Count			LINER	Soil Classification
			1	2	3		
0.5							6" TOPSOIL MIXTURE SILT
1	SUR	SS	4	5	—		
5	SUR	SS	3	4	—		BROWN
	SUR	SS	2	3	—		
10	SUR	SS	2	2	—		FINE TO
15	SUR	SS	6	15	—		MEDIUM
20	SUR	SS	3	5	—		SAND
25	SUR	SS	5	9	—		
							END OF BORING 25.5 FT.
30							
35							

Gosling Czubak

Associates



Engineers
Surveyors

Gosling Czubak Associates, P.C.
525 West Fourteenth
Traverse City, MI 49684
616 946-9191

PROJECT Four Star
LOCATION Cadillac, Michigan
CLIENT Ron Sondee

DATE STARTED 5/16/83
DATE COMPLETED 5/16/83
DRILLER BRK HELPER BB & JCM

BORING NO. SB #11
Ground Surface Elev. 1297.7
Plugging Method Existing Material

Weather Cloudy & Calm 50°F
Water Data 29%± below ground
Job No. 83072.01

Depth	Sample	Sampling Method	PENETR.: Blow Count			LINER	Soil Classification
			1	2	3		
0.5							6" TOPSOIL
1.5							12" BROWN MEDIUM SAND
2.1	SUR	SS	6	7	—		6" BLACK SAND
							REDDISH-BROWN FINE TO MEDIUM SAND
5	SUR	SS	1	1	1		
10	SUR	SS	3	4	—		BROWN
15	SUR	SS	2	4	—		FINE TO
20	SUR	SS	3	5	—		MEDIUM
25	SUR	SS	5	5	—		SAND
29.5	SUR	SS	5	7	—		
30							END OF BORING 30.5 FT.
35							

Gosling Czubak

Associates



Engineers
Surveyors

Gosling Czubak Associates, P.C.
525 West Fourteenth
Traverse City, MI 49684
616 946-9191

PROJECT Four Star
LOCATION Cadillac, Michigan
CLIENT Ron Sondee

DATE STARTED 5/16/83
DATE COMPLETED 5/16/83
DRILLER BRK HELPER BB & JCM

BORING NO. SB #10
Ground Surface Elev. 1297.5
Plugging Method Existing Material

Weather Cloudy & Calm 45°F
Water Data 29 1/2' ± below ground
Job No. 83072.01

Depth	Sample	Sampling Method	PENETR.: Blow Count			LINER	Soil Classification
			1	2	3		
0.3							6" TOP SOIL
0.5	SUR	AUGER					12" BROWN MEDIUM SAND
0	SUR	SS	3	2			6" BLACK SAND
5	SUR	SS	1	2			REDDISH-BROWN FINE TO MEDIUM SAND
	SUR	SS	3	4			
10	SUR	SS	3	5			BROWN
15	SUR	SS	4	5			FINE TO
20	SUR	SS	4	7			MEDIUM
25	SUR	SS	5	8			SAND
29.5	SUR	SS	7	14			
30							END OF BORING 30.5 FT.
35							

Gosling Czubak
Associates



Engineers
Surveyors

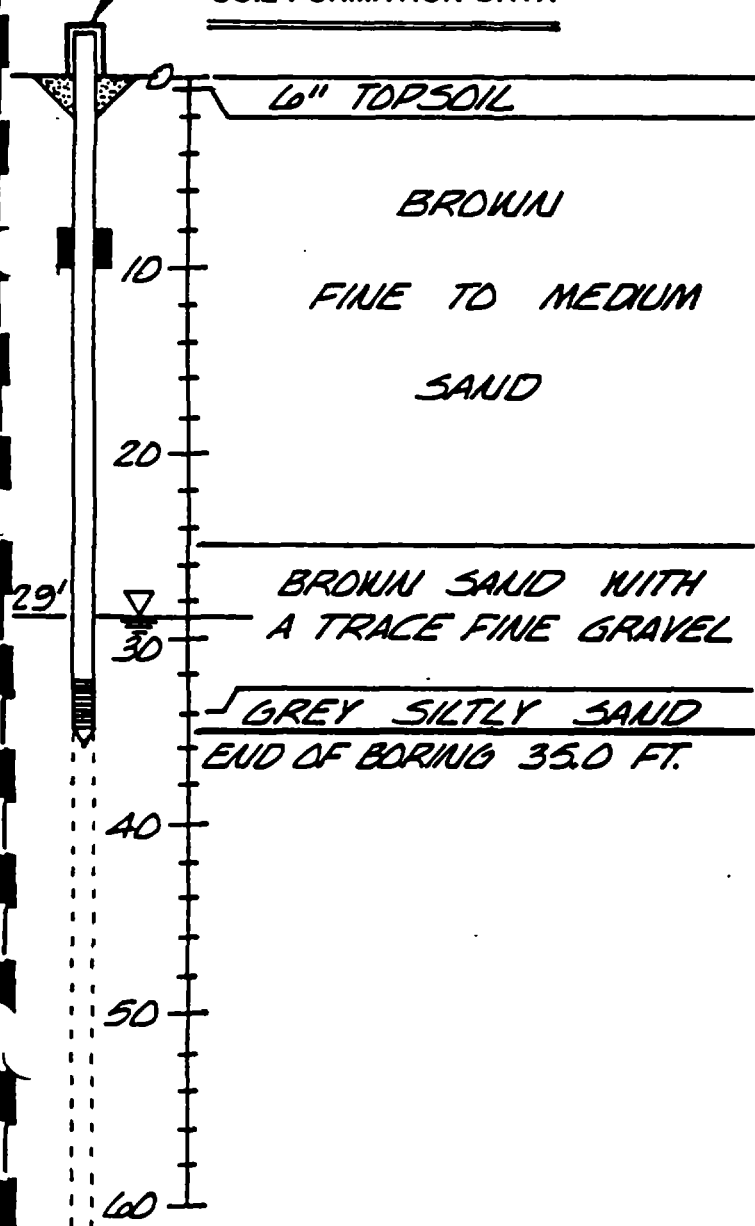
Gosling Czubak Associates, P.C.
525 West Fourteenth
Traverse City, MI 49684
616 946-9191

MONITOR WELL LOG

Project FOUR STAR
Location CADILLAC, MICHIGAN.
Client RON SONDEE
Well Designation M.W. #10
Date Installed MAY 16, 1983

Development Method PUMPING
Ground Elevation 1295.7
Top of Casing Elev. 1297.78
Driller BRK Helper BB & JCM
Job Number 83072.01

CASING PROTECTION SOIL FORMATION DATA



CASING DATA

Type 2" GALVANIZED
Length 34.5 FT.

SCREEN DATA

Type STAINLESS STEEL GAUZE
Size 2" DIA., 60 GAUGE
Length 30"

SAMPLING RECORD

Date MAY 16, 1983
Comments PUMPED W/ JET



BENTONITE SEAL

CONCRETE

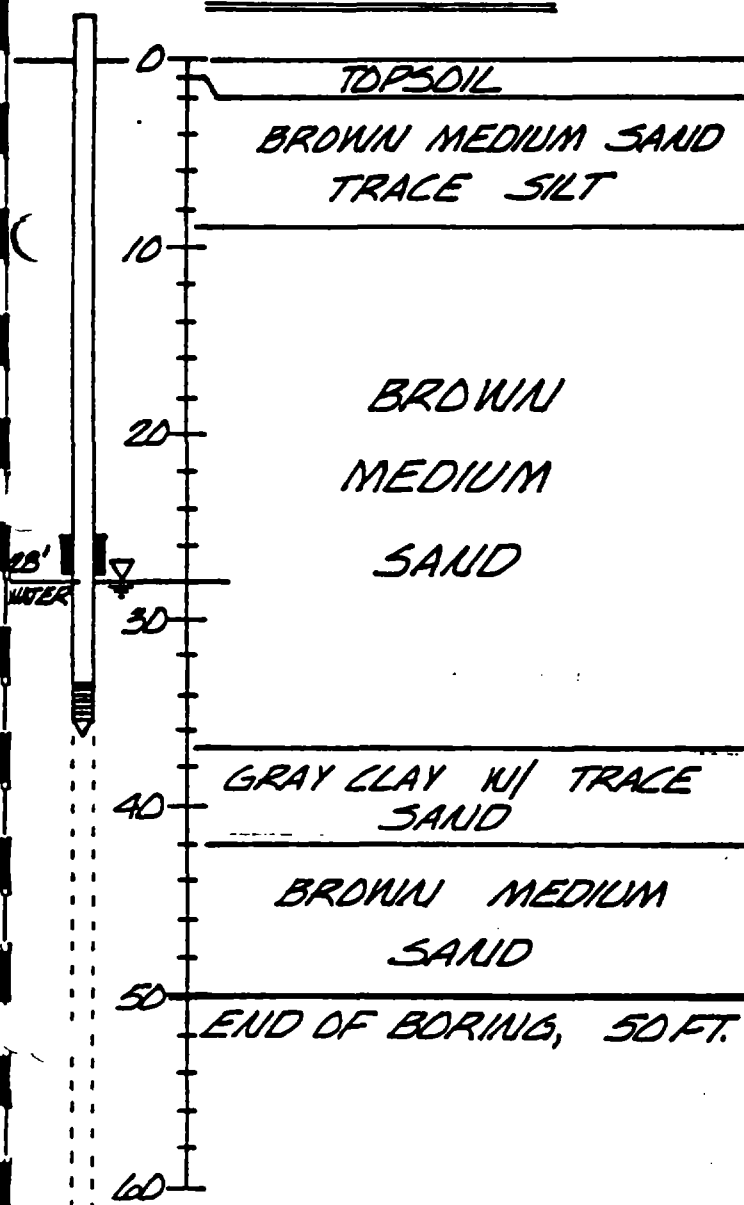


MONITOR WELL LOG

Project FOUR STAR
Location CADILLAC, MICHIGAN
Client RON SQUIDEE
Well Designation M.W. #10
Date Installed JULY 5, 1983

Development Method PUMPING
Ground Elevation 1297.8
Top of Casing Elev. 1299.62
Driller KUENZER Helper BUFKA
Job Number 83072.01

SOIL FORMATION DATA



CASING DATA

Type 2" DIA. GALVANIZED STEEL
Length 35.5 FT.

SCREEN DATA

Type STAINLESS STEEL MESH,
DRIVE POINT
Size NO. 60 MESH
Length 2 FT.

SAMPLING RECORD

Date JULY 5, 1983, B. KUENZER
Comments _____

■ BENTONITE SEAL

Gosling Czubak
Associates



Engineers
Surveyors

Gosling Czubak Associates, PC.
525 West Fourteenth
Traverse City, MI 49684
616 946-9191

MONITOR WELL LOG

Project FOUR STAR
Location CADILLAC, MICHIGAN
Client RON SONDEE
Well Designation M.W.#8
Date Installed JULY 5, 1983

Development Method PUMPING
Ground Elevation 1299.0
Top of Casing Elev. 1301.45
Driller KUENZER Helper BUFKA
Job Number 83072.01

SOIL FORMATION DATA

0
SANDY TOPSOIL
BROWN MEDIUM SAND
WITH TRACE OF SILT

10
20
BROWN
MEDIUM
SAND

30
40
50
60
END OF BORING, 50 FT.

CASING DATA

Type 2" DIA. GALVANIZED STEEL
Length 42 FT.

SCREEN DATA

Type STAINLESS STEEL MESH,
DRIVE POINT
Size NO. 60 MESH
Length 2 FT.

SAMPLING RECORD

Date JULY 5 1983, B. KUENZER
Comments _____

■ BENTONITE SEAL

Gosling Czubak

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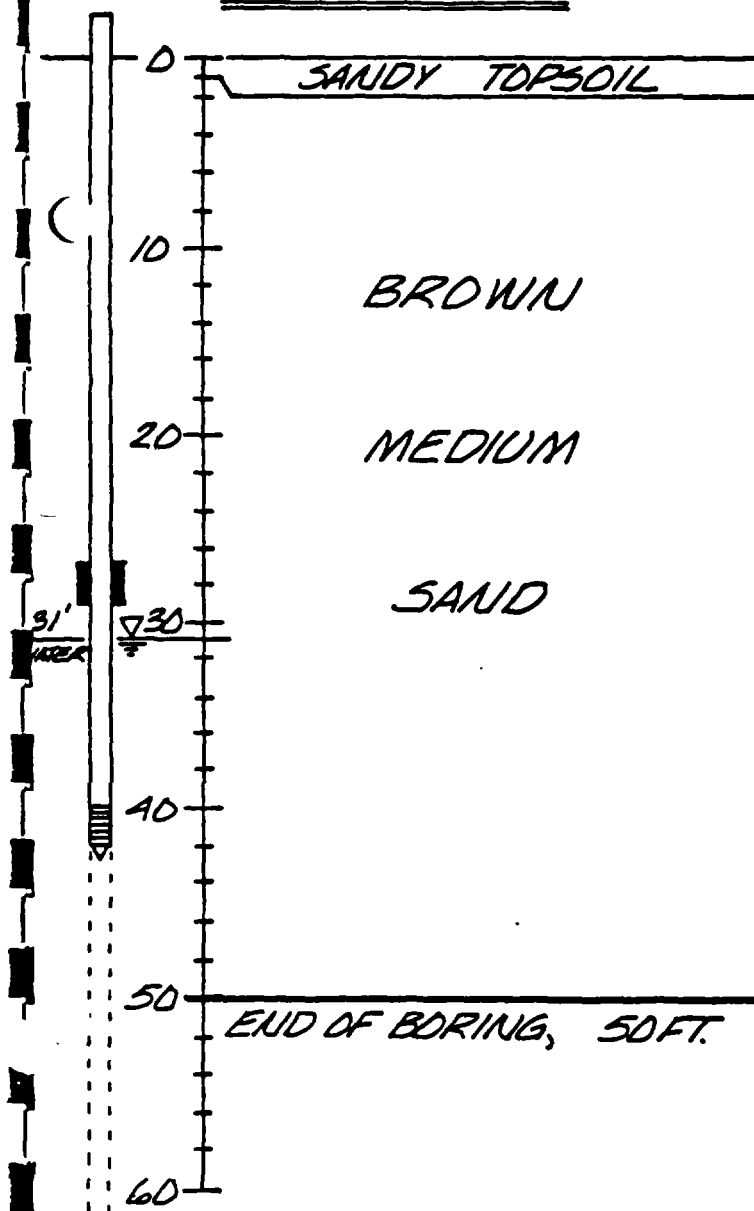
Gosling Czubak Associates, PC.
525 West Fourteenth
Traverse City, MI 49684
616 946-9191

MONITOR WELL LOG

Project FOUR STAR
Location CADILLAC, MICHIGAN
Client RON SONDEE
Well Designation M. W. # 7
Date Installed JULY 5, 1983

Development Method PUMPING
Ground Elevation 1299.05
Top of Casing Elev. 1301.39
Driller KUENZER Helper BUFKA
Job Number 83072.01

SOIL FORMATION DATA



CASING DATA

Type 2" ϕ GALVANIZED STEEL
Length 42 FT.

SCREEN DATA

Type STAINLESS STEEL MESH,
DRIVE POINT
Size NO. 60 MESH
Length 2 FT.

SAMPLING RECORD

Date JULY 5 1983, B. KUENZER
Comments _____

 BENTONITE SEAL

Gosling Czubak
Associates



Engineers
Surveyors

Gosling Czubak Associates, P.C.
525 West Fourteenth
Traverse City, MI 49684
616 946-9191

MONITOR WELL LOG

Project FOUR STAR
Location CADILLAC, MICHIGAN
Client RON SAUDEE
Well Designation M.W. #9
Date Installed JULY 5, 1983

Development Method PUMPING
Ground Elevation 1298.8
Top of Casing Elev. 1301.20
Driller KUENZER Helper BUFKA
Job Number 83072.01

SOIL FORMATION DATA

0
SAUDY TOPSOIL
BROWN MEDIUM SAND
WITH TRACE OF SILT

10
20
BROWN
MEDIUM
SAND

30
31
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END OF BORING, 50 FT.
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CASING DATA

Type 2" DIA. GALVANIZED STEEL
Length 42 FT.

SCREEN DATA

Type STAINLESS STEEL MESH,
DRIVE POINT.
Size NO. 60 MESH
Length 2 FT.

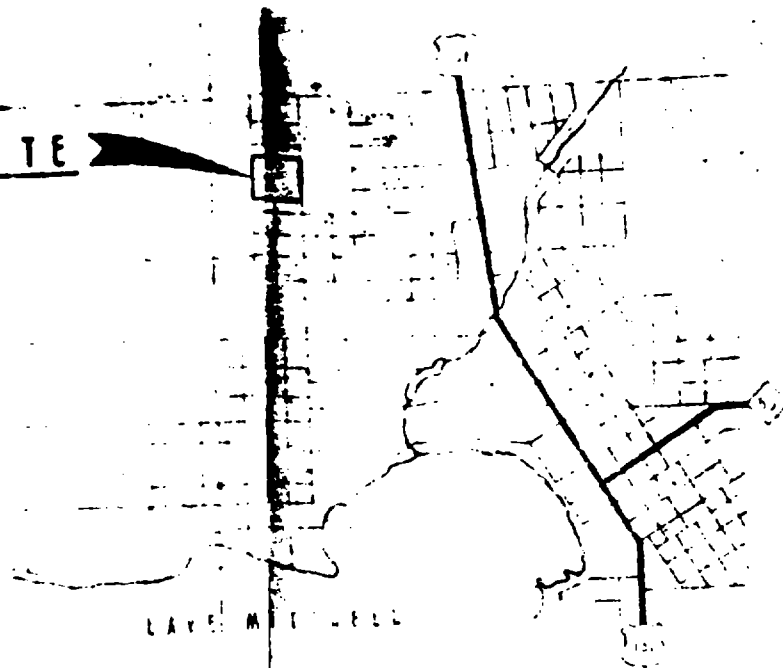
SAMPLING RECORD

Date JULY 5, 1983, B. KUENZER
Comments _____

 BENTONITE SEAL

APPENDIX B
GROUNDWATER CONTOUR
AND
CONTAMINANT RISK MAP

SITE



VICINITY MAP
CITY OF CADILLAC

LEGEND

WATER

ROAD

RAILROAD

UNIMPROVED ROAD

UNIMPROVED ROAD

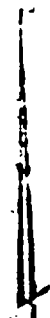
TRICHLOROETHYLENE PLUME

IN PARTS PER MILLION

GROUNDWATER CONTOUR

GROUNDWATER FLOW

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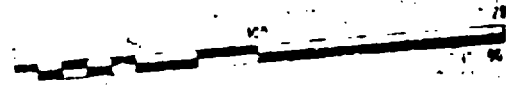
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LEGEND

05-0-442

ETHYLENE PLUME
IN PARTS PER MILLION
GROUNDWATER CONTOUR
GROUNDWATER FLOW

N



Gosling Czubak ASSOCIATES



Engineers
Surveyors

10000 Lakeside Avenue
Livonia, Michigan 48150
Phone: 481-4444
Telex: 254111

CONTAMINANT PLUME

FORMER FOOD STAR PLAN

12345 1ST AVE NEAR 220 CO. MICHIGAN SEE

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WAG. 2

8042-442

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PLUME CONFIGURATION

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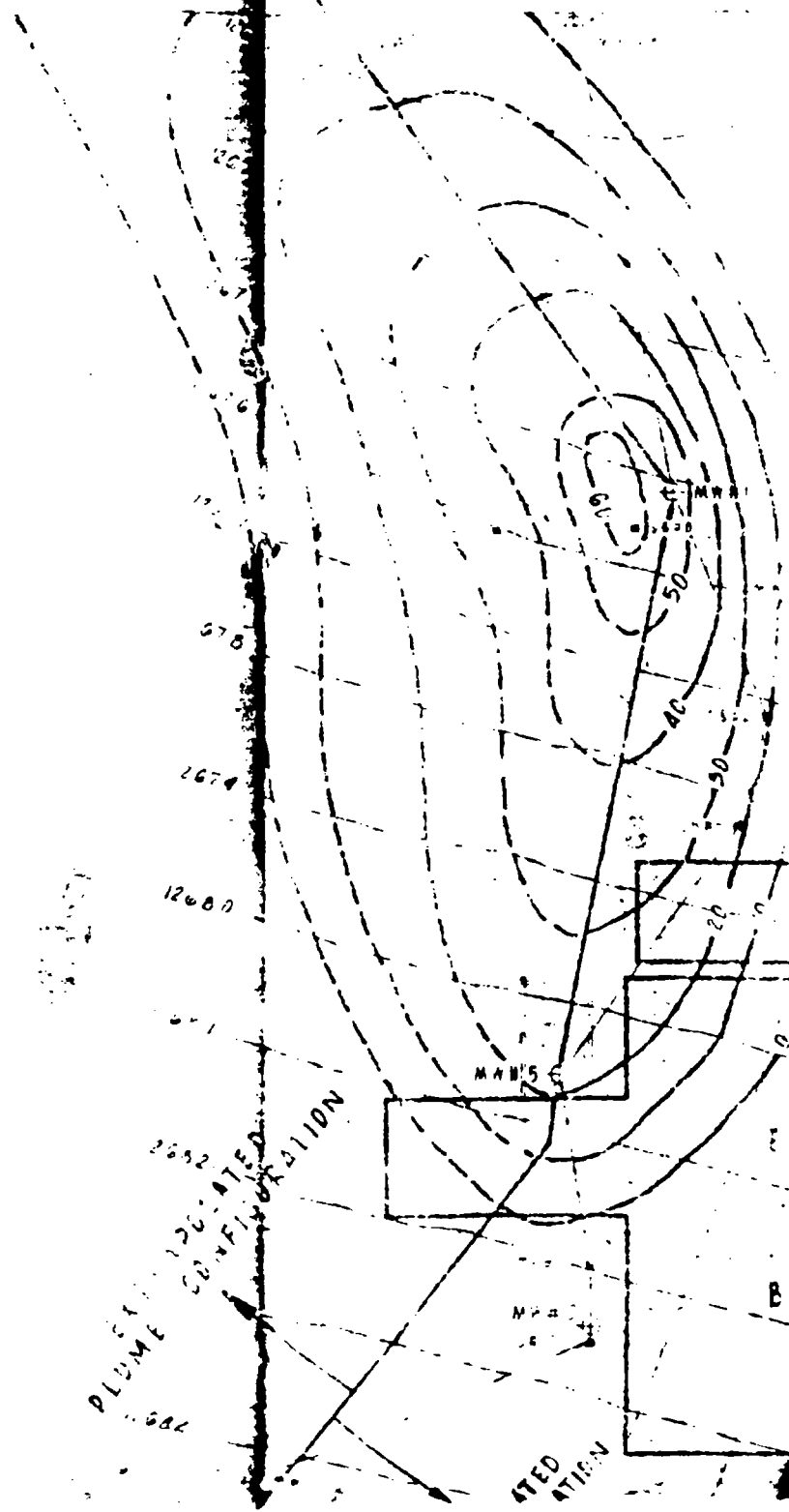
570

MAR 5

MAR 2

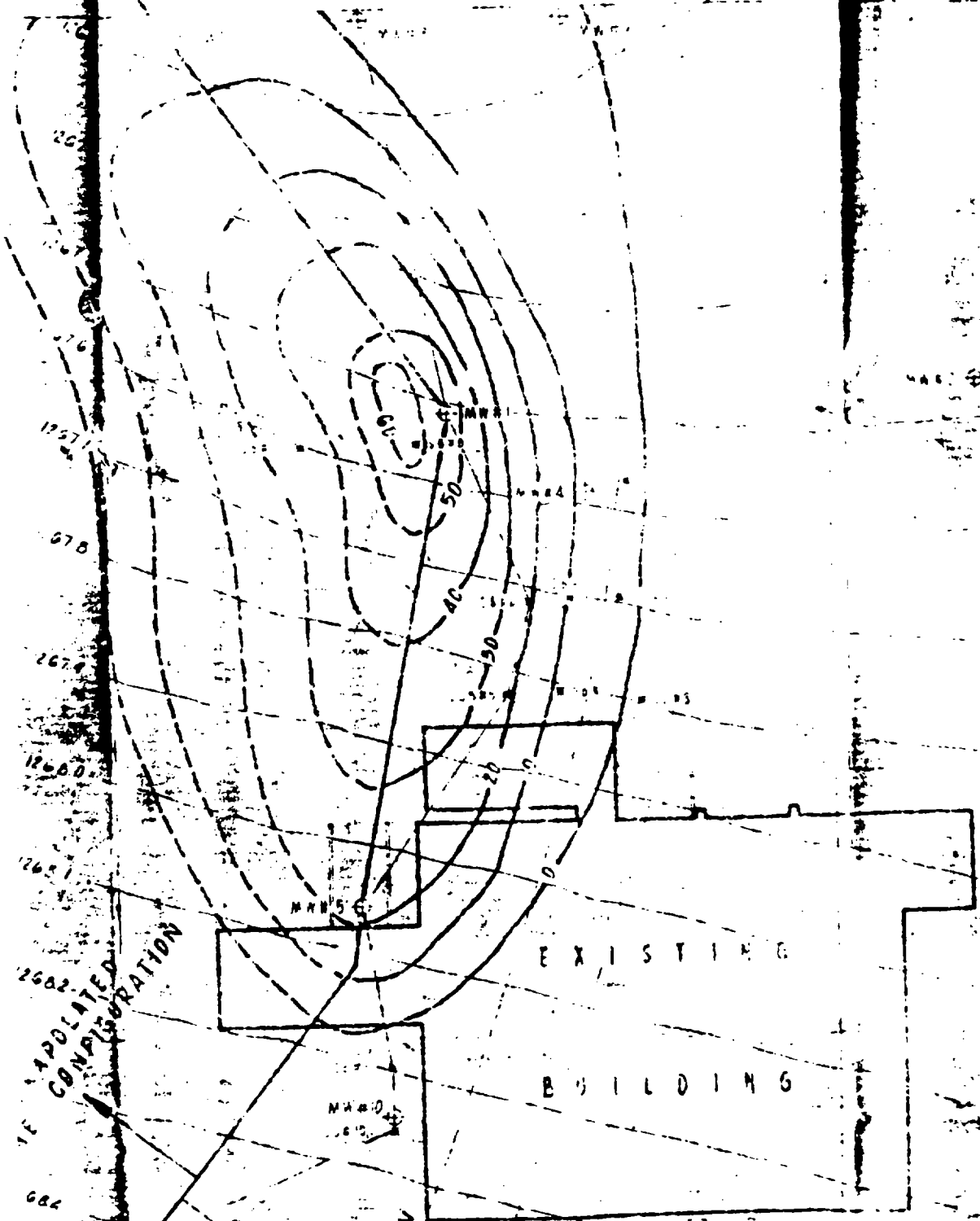
STATION

B



EXISTING

BUILDING



EXISTING

BUILDING

ADOLATED
CONSPICUOUS

LATED
RATION

12684
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EXTRAPOLATED
PLUME CONFIGURATION

INTERPOLATED
PLUME CONFIGURATION

TRICHLOROETHYLENE DL

1-16-62-PC-11-10

DO NOT REMOVE
EX FILE NUMBER
FILE 19999

WATER LEVEL DATA		
STATION NO.	ELEVATION (FSD)	STAGE (FSD)
100	126.00	126.00
101	126.00	126.00
102	126.00	126.00
103	126.00	126.00
104	126.00	126.00
105	126.00	126.00
106	126.00	126.00
107	126.00	126.00
108	126.00	126.00
109	126.00	126.00
110	126.00	126.00

WELL, VALIDATING WATER
LEVEL TO 4' FROM SURFACE

GRAND JURY

Gosling Cubak Associate

100

1998

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SECRET

5000. 5-11-1962

DATE: 6.12.2017

FORMER - COOR - STAR - 100A

HARING IN WEXFORD CO. NIGHTINGALE

SITE



100 METERS

PROXIMITY MAP

1:50,000 SCALE

1:50,000 SCALE

BENCHMARK

NO. 10-1000-10
IN 1000 HYDRAULIC
DATE 1944

REMARKS

COLLECTED

STATION

DATE

TIME

CHANGING

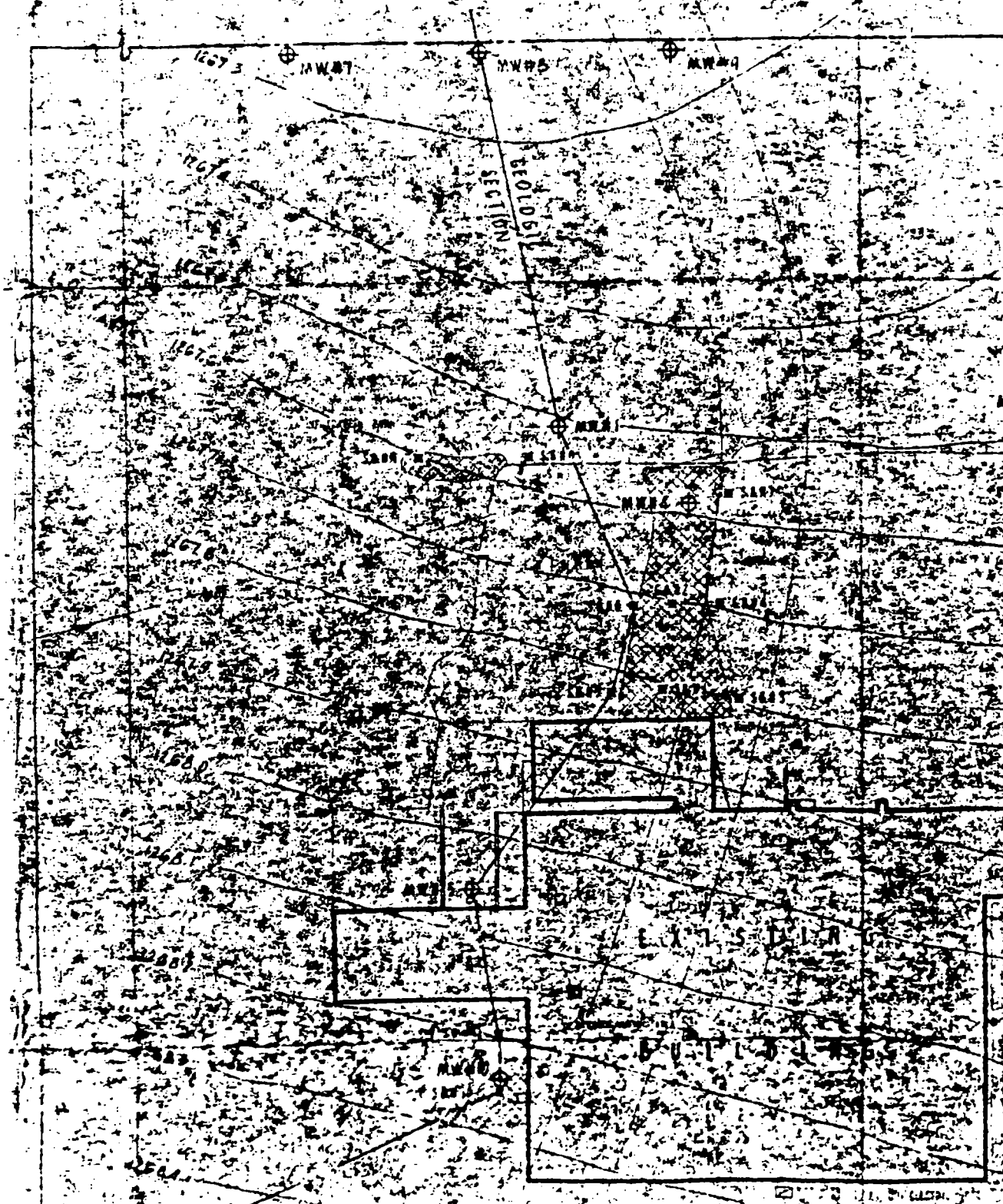
WATER

WATER LEVEL DATA

STATION	DATE	TIME	WATER LEVEL (FEET)
1	1944	12:00	1267.45
2	1944	12:05	1267.44
3	1944	12:10	1267.50
4	1944	12:15	1267.63
5	1944	12:20	1268.32
6	1944	12:25	1268.24
7	1944	12:30	1267.56
8	1944	12:35	1267.74
9	1944	12:40	1267.95

NO. 10-1000-10
IN 1000 HYDRAULIC
DATE 1944

B U I L D I N G



GEODESIC SECTION

1.40

0.50

0.30

0.20

0.10

0.05

0.02

0.01

0.005

0.002

0.001

0.0005

0.0002

0.0001

0.00005

0.00002

0.00001

0.000005

0.000002

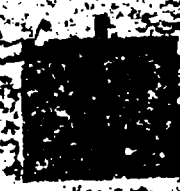
0.000001

NORTH PARK PLAT

THIRTEENTH STREET



LIBRARY OF THE
CITY OF NEW YORK



SIXTH AVENUE

C MUNICIPAL
LL FIELD

Z

SCALE 1"=100'

FILE 4411

FORMER FOUR
STAR PLANT

SEVENTH STREET

CADIL

SCALE = 100

FIVE ALLEY

SEVENTH STREET

SEVENTH STREET

CADILLAC MUNICIPAL
WELL FIELD

TH AVENUE

Continued on next page

Costing - 1954 - 1955

NOTICE SECTION